Statistical Mechanics Rediscovered through the Lens of the Entropy Maximization Principle

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Part I

Statistical Derivation of Ensembles

1 Entropy Maximization Principle

The conventional approach and its limitation. The conventional introduction to statistical mechanics rests on four notions:

- A microstate is a complete description of every microscopic parameter of the system.
- In an isolated system, all microstates are equally likely.
- A macrostate is a group of microstates that share the same macroscopic observables.
- Macrostates that contain more microstates are observed more often, so the observed macrostate is the most probable one.

Boltzmann, Gibbs and other founders of thermodynamics developed this approach. The approach is physically intuitive. However, an explanation of why a system settles into the most probable macrostate requires ergodicity and dynamical arguments, which are difficult to make rigorous.

The MaxEnt alternative. This document follows Jaynes in treating equilibrium statistical mechanics as a constrained optimization problem, in which the entropy is maximized subject to what is known macroscopically about the system. This approach is called the Entropy Maximization Principle (MaxEnt).

1.1 Gibbs–Shannon Entropy

From "disorder" to ignorance. The meaning of entropy has long been debated. The popular description of entropy as "disorder" is not a definition, because the description gives no quantitative measure of disorder. Clausius, Boltzmann and other founders regarded entropy as a driving force of thermodynamic processes, which left open the question of which physical quantity entropy encodes. Jaynes instead argued that entropy is not a property of the body itself but a measure of the observer's ignorance about the body. The Gibbs–Shannon entropy formalizes this idea for Ω events that occur with probabilities $p(i)$:

$$\boxed{S(p) = -\sum_{i=1}^{\Omega} p(i) \ln p(i).} \tag{1.1}$$

Why the Gibbs–Shannon entropy measures ignorance. The term $-\ln p(i)$, called the surprisal of event i, measures the ignorance about that event:

- A certain event has $p(i) = 1$, so the surprisal is $-\ln 1 = 0$.
- A rare event has a small $p(i)$, so the surprisal is large.

A direct sum of $-\ln p(i)$ would fail, because knowing that an event will occur and knowing that the event will not occur both mean low ignorance. Each term is therefore weighted by $p(i)$. Rare events seldom occur, and near-certain events are already expected, so both contribute little to the total. Thus, S is the expected surprisal, $\langle -\ln p \rangle$.

1.2 Entropy Maximization

The principle of maximum ignorance. Jaynes' second idea is to choose the distribution that maximizes the ignorance, that is, the Gibbs–Shannon entropy $S(p)$, subject to the known constraints whenever the underlying mechanism is unknown. A lower-entropy distribution would introduce an unexplained bias toward some events. For example, the two outcomes of a coin flip should have equal probabilities unless a constraint, such as a loaded coin, states otherwise.

Formulating the constrained optimization. For simplicity, p_i replaces $p(i)$ from now on, because only one probability distribution appears. Each known constraint is written as $g_a(p) = 0$. For a constraint on the expectation value of an observable that takes the value $g_a(i)$ in event i,

$$g_a(p) = \sum_i p_i \, g_a(i) - \langle g_a \rangle .$$

Lagrange multipliers remove the constraints through the Lagrangian

$$\boxed{\mathcal{L}(p) = -\sum_i p_i \ln p_i - \alpha \left(\sum_i p_i - 1 \right) - \sum_a \lambda_a \, g_a(p),} \tag{1.2}$$

where α enforces normalization and each λ_a enforces one constraint. The maximizing distribution satisfies

$$\boxed{\frac{\partial \mathcal{L}}{\partial p_i} = 0, \qquad \frac{\partial \mathcal{L}}{\partial \alpha} = 0, \qquad \frac{\partial \mathcal{L}}{\partial \lambda_a} = 0.} \tag{1.3}$$

The last two conditions reproduce normalization and the constraints. The first condition determines p_i.

The general MaxEnt distribution. For expectation-value constraints, $\partial g_a / \partial p_i = g_a(i)$. The first condition in Eq. (1.3) then gives

$$-\ln p_i - 1 - \alpha - \sum_a \lambda_a \, g_a(i) = 0,$$

$$p_i = e^{-1-\alpha} \exp\Big(-\sum_a \lambda_a \, g_a(i) \Big).$$

Normalization fixes α and yields

$$\boxed{p_i = \frac{1}{Z} \exp\Big(-\sum_a \lambda_a \, g_a(i) \Big), \qquad Z = \sum_i \exp\Big(-\sum_a \lambda_a \, g_a(i) \Big).} \tag{1.4}$$

Since S is strictly concave in p and the constraints are linear, this stationary point is the unique maximum.

2 The Microcanonical Ensemble

The Boltzmann constant in the entropy. In thermodynamics, the Gibbs–Shannon entropy includes the Boltzmann constant k_B, which sets the entropy units:

$$\boxed{S(p) = -k_B \sum_{i=1}^{\Omega} p_i \ln p_i.} \tag{2.1}$$

For simplicity, k_B is set to 1 during derivations and restored in the final results.

From normalization alone to Boltzmann's entropy. The simplest case has normalization, $\sum_i p_i = 1$, as the only constraint. This case describes the microcanonical ensemble, an isolated system with fixed energy and other fixed macroscopic constraints. The energy is not imposed through a Lagrange multiplier. Instead, the energy restricts the sample space to the microstates with energy U, in practice a thin shell $[U, U + \delta U]$. The stationarity conditions of Eq. (1.3) are then

$$\frac{\partial \mathcal{L}}{\partial p_i} = -\ln p_i - 1 - \alpha = 0, \qquad \frac{\partial \mathcal{L}}{\partial \alpha} = -\Big(\sum_i p_i - 1\Big) = 0.$$

The first condition gives $p_i = e^{-1-\alpha}$, which does not depend on i. Normalization then requires $\Omega\, e^{-1-\alpha} = 1$, so

$$\boxed{p_i = \frac{1}{\Omega}.} \tag{2.2}$$

The entropy maximum under normalization alone is therefore a uniform distribution, in which all accessible microstates are equally probable. The conventional approach postulates this property, whereas MaxEnt derives the property and thereby recovers Boltzmann's premise for the microcanonical ensemble. Substitution of $p_i = 1/\Omega$ into the entropy gives

$$S = -\sum_{i=1}^{\Omega} \frac{1}{\Omega} \ln \frac{1}{\Omega} = \ln \Omega,$$

and, with k_B restored, Boltzmann's microcanonical entropy is

$$\boxed{S = k_B \ln \Omega.} \tag{2.3}$$

Dependence of the entropy on the macroscopic constraints. The number of accessible microstates, Ω, is also called the microcanonical partition function. This number depends on the Hamiltonian $\mathcal{H}$ of the system and on the macroscopic constraints. The volume V, the particle number N and the internal energy U set the accessible microstates, so the entropy is a function of these three variables:

$$\boxed{S = S(N, V, U).} \tag{2.4}$$

Specified values of N, V and U therefore determine the equilibrium entropy uniquely.

3 The Canonical Ensemble

Constraining the average of an observable. The microcanonical ensemble does not distinguish between accessible microstates, so only their number matters. However, realistic systems obey constraints imposed by measurement. Suppose microstate i has a definite value a_i of an observable A, and the measured average is $\bar{A}$. The expectation value is

$$\boxed{\langle A\rangle \equiv \sum_i p_i a_i,} \tag{3.1}$$

and the constraint reads $\langle A\rangle = \bar{A}$. Here, $\langle A\rangle$ denotes the operation that extracts the mean of the observable A, and $\bar{A}$ denotes the numerical value assigned to that mean. Since only one constraint is present, the multiplier λ_a is renamed β. The Lagrangian now contains the multiplier α for normalization and the multiplier β for the constraint on $\langle A\rangle$. The stationarity conditions of this Lagrangian are

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial p_i} &= -\ln p_i - 1 - \alpha - \beta a_i = 0,\\
\frac{\partial \mathcal{L}}{\partial \alpha} &= -\Big(\sum_i p_i - 1\Big) = 0,\\
\frac{\partial \mathcal{L}}{\partial \beta} &= -\Big(\sum_i a_i p_i - \bar{A}\Big) = 0.
\end{aligned}$$

The Boltzmann distribution and the partition function. The first condition gives $p_i = e^{-1-\alpha-\beta a_i}$. Substitution of this result into the normalization condition and the constraint yields

$$e^{-1-\alpha}\sum_i e^{-\beta a_i} = 1, \qquad e^{-1-\alpha}\sum_i a_i e^{-\beta a_i} = \bar{A}.$$

The first equation determines α in terms of β and leaves a single free parameter. The canonical partition function is defined as

$$\boxed{Z(\beta) \equiv \sum_i e^{-\beta a_i},} \tag{3.2}$$

so the probability distribution becomes

$$\boxed{p_i = \frac{e^{-\beta a_i}}{Z(\beta)}.} \tag{3.3}$$

This result is the Boltzmann distribution. For $\beta > 0$, states with larger a_i are less probable. The remaining constraint, $\langle A\rangle(\beta) = \bar{A}$, determines β.

Uniqueness of the multiplier. The uniqueness of β follows from the mean expressed through Z:

$$\langle A\rangle = \frac{1}{Z}\sum_i a_i e^{-\beta a_i} = -\frac{\partial \ln Z}{\partial \beta}.$$

Hence

$$\boxed{\langle A\rangle = -\frac{\partial \ln Z}{\partial \beta}.} \tag{3.4}$$

With $\mathrm{Var}(A) \equiv \langle A^2\rangle - \langle A\rangle^2$, the derivative of the mean is

$$\begin{aligned}\frac{\partial\langle A\rangle}{\partial\beta} &= \frac{\partial}{\partial\beta}\left(\frac{\sum_i a_i e^{-\beta a_i}}{Z}\right)\\ &= -\frac{\sum_i a_i^2 e^{-\beta a_i}}{Z} + \frac{1}{Z^2}\left(\sum_i a_i e^{-\beta a_i}\right)^2\\ &= -\langle A^2\rangle + \langle A\rangle^2 = -\mathrm{Var}(A),\end{aligned}$$

so

$$\boxed{\frac{\partial\langle A\rangle}{\partial\beta} = -\mathrm{Var}(A).} \tag{3.5}$$

Since $\mathrm{Var}(A) \geq 0$, $\langle A\rangle(\beta)$ is monotonically non-increasing. The mean is strictly decreasing whenever A takes more than one value over the accessible microstates. Thus, for target values in the interior of the achievable range, $\langle A\rangle$ determines β uniquely.

Entropy of the canonical distribution and the meaning of β. The entropy of the resulting distribution, with k_B omitted during the derivation, is

$$\begin{aligned}S &= -\sum_i p_i \ln p_i = -\sum_i p_i\left(-\beta a_i - \ln Z\right)\\ &= \beta\sum_i a_i p_i + \ln Z\sum_i p_i = \beta\langle A\rangle + \ln Z.\end{aligned}$$

With k_B restored,

$$\boxed{S = k_B\left(\beta\langle A\rangle + \ln Z\right).} \tag{3.6}$$

Although S depends on $\bar{A}$ both explicitly and implicitly through $\beta(\bar{A})$, the derivative of S reduces to β:

$$\begin{aligned}\frac{\partial S}{\partial\langle A\rangle} &= \beta + \langle A\rangle\frac{d\beta}{d\langle A\rangle} + \frac{\partial \ln Z}{\partial\beta}\frac{d\beta}{d\langle A\rangle}\\ &= \beta + \langle A\rangle\frac{d\beta}{d\langle A\rangle} - \langle A\rangle\frac{d\beta}{d\langle A\rangle} = \beta.\end{aligned}$$

With k_B restored, this result gives the meaning of the multiplier:

$$\boxed{\beta = \frac{1}{k_B}\frac{\partial S}{\partial\langle A\rangle}.} \tag{3.7}$$

Concavity of the entropy. A function f is concave on an interval I if, for all $x, y \in I$ and all $t \in [0,1]$,

$$\boxed{f\big(tx + (1-t)y\big) \;\geq\; t\,f(x) + (1-t)\,f(y).} \tag{3.8}$$

Geometrically, every chord lies on or below the graph. The function is strictly concave if the inequality is strict for $x \neq y$ and $t \in (0,1)$. For a twice-differentiable f, concavity is equivalent to $f'' \leq 0$, and $f'' < 0$ guarantees strict concavity. For S as a function of $\langle A \rangle$, the inverse-function rule and $\partial\langle A\rangle/\partial\beta = -\mathrm{Var}(A)$ give

$$\frac{\partial^2 S}{\partial \langle A\rangle^2} = \frac{d\beta}{d\langle A\rangle} = \left(\frac{d\langle A\rangle}{d\beta}\right)^{-1} = -\frac{1}{\mathrm{Var}(A)}.$$

With k_B restored,

$$\boxed{\frac{\partial^2 S}{\partial \langle A\rangle^2} = -\frac{k_B}{\mathrm{Var}(A)} < 0.} \tag{3.9}$$

The entropy is therefore strictly concave in $\langle A \rangle$ whenever A takes more than one value over the accessible microstates. This concavity is the monotonic decrease of $\langle A\rangle(\beta)$ viewed from the side of S, with each value of $\langle A\rangle$ corresponding to exactly one β.

Specialization to the internal energy. Since the observable A was never specified, the derivation applies to any observable with a constrained expectation value. The canonical ensemble conventionally fixes the volume V and the particle number N and constrains the internal energy. From here on, $A = U$, and U_i replaces a_i. The results become

$$\boxed{\begin{aligned} Z(\beta) &= \sum_i e^{-\beta U_i}, \quad p_i = \frac{e^{-\beta U_i}}{Z(\beta)}, \\ \langle U\rangle &= -\frac{\partial \ln Z}{\partial \beta}, \qquad S = k_B\left(\beta\langle U\rangle + \ln Z\right). \end{aligned}} \tag{3.10}$$

Section 3.1 identifies β with the temperature. The end of this section explains why the particle number or the volume cannot be constrained alone.

Recovering the microcanonical ensemble. If every accessible microstate has the same energy U, the canonical ensemble reduces to the microcanonical ensemble. All the weights $e^{-\beta U}$ are identical, so the distribution is uniform and the mean energy equals U exactly. Then

$$\begin{aligned} Z &= \sum_{i=1}^{\Omega} e^{-\beta U} = \Omega\, e^{-\beta U}, \\ S &= k_B\left(\beta U + \ln\Omega - \beta U\right) = k_B \ln\Omega. \end{aligned}$$

3.1 First and Second Laws of Thermodynamics — Meaning of the Multipliers

The multipliers need thermodynamics for their meaning. The MaxEnt formalism does not by itself provide the physical interpretation or the numerical calibration of the multipliers. In particular, the identification

$$\boxed{\beta = \frac{1}{k_B}\frac{\partial S}{\partial \langle U\rangle} = \frac{1}{k_B T}} \tag{3.11}$$

cannot be derived from MaxEnt alone. This identification is an empirical correspondence established by classical thermodynamics. The physical meaning of the multipliers therefore comes from the thermodynamic laws.

From the first and second laws to the fundamental relation. The first law expresses energy conservation:

$$\boxed{dU = \delta Q - \delta W + \mu\, dN,} \tag{3.12}$$

where

- $\delta W = P\, dV$ is the work done *by* the system.
- The symbols δQ and δW emphasize that heat and work are not state functions, because their values depend on the path between two states.
- The chemical potential μ is the change in internal energy per added particle at fixed entropy and volume.

The second law, in the form of the Clausius inequality, states

$$\boxed{\delta Q \leq T\, dS,} \tag{3.13}$$

with equality for reversible processes. For a reversible process, the two laws combine to give

$$\boxed{dU = T\, dS - P\, dV + \mu\, dN,} \tag{3.14}$$

and therefore

$$\boxed{\left.\frac{\partial U}{\partial S}\right|_{V,N} = T, \qquad \left.\frac{\partial U}{\partial V}\right|_{S,N} = -P, \qquad \left.\frac{\partial U}{\partial N}\right|_{S,V} = \mu.} \tag{3.15}$$

The equivalent entropy form of Eq. (3.14) is

$$\boxed{dS = \frac{1}{T}\, dU + \frac{P}{T}\, dV - \frac{\mu}{T}\, dN,} \tag{3.16}$$

and therefore

$$\boxed{\left.\frac{\partial S}{\partial U}\right|_{V,N} = \frac{1}{T}, \qquad \left.\frac{\partial S}{\partial V}\right|_{U,N} = \frac{P}{T}, \qquad \left.\frac{\partial S}{\partial N}\right|_{U,V} = -\frac{\mu}{T}.} \tag{3.17}$$

3.2 Extensive and Intensive Variables

Extensive and intensive variables and their conjugate pairs.

- A quantity X is extensive if, for two weakly coupled subsystems, $X_{\text{total}} = X_1 + X_2$. Equivalently, for λ identical, non-interacting copies, $X_\lambda = \lambda X_1$.

- A quantity Y is intensive if Y does not scale with the number of copies, $Y_\lambda = Y_1$.
- Energy, particle number, volume and entropy are extensive, whereas temperature, pressure and chemical potential are intensive.

This distinction explains why the Lagrange multipliers associated with extensive constraints are intensive. For any observable A with constrained average $\langle A \rangle$, the multiplier is

$$\beta_A \equiv \frac{1}{k_B} \frac{\partial S}{\partial \langle A \rangle}.$$

Because $\langle A \rangle$ and S are both extensive, their scaling factors cancel in the derivative, so β_A is intensive. Thermodynamic variables therefore come in conjugate pairs, each consisting of an extensive variable and its intensive conjugate.

Additivity of entropy follows from statistical independence. Here, the superscripts (1) and (2) also label the distributions of the two systems, $p_i^{(1)}$ and $p_j^{(2)}$. The partition function factorizes because the joint distribution factorizes, $p_{ij} = p_i^{(1)} p_j^{(2)}$. The entropy is then additive:

$$S_{1+2} = -\sum_{i,j} p_i^{(1)} p_j^{(2)} \ln\!\left(p_i^{(1)} p_j^{(2)}\right) = S_1 + S_2.$$

Additivity requires statistical independence but not equilibrium, so the two systems may have different multipliers β_1 and β_2:

$$\boxed{S_{1+2} = k_B \left(\beta_1 \langle U_1 \rangle + \ln Z_1 + \beta_2 \langle U_2 \rangle + \ln Z_2\right).} \tag{3.18}$$

Only when $\beta_1 = \beta_2 = \beta$, that is, when the two systems are in thermal equilibrium, does the joint system become canonical with a single multiplier:

$$\boxed{S_{1+2} = k_B \left[\beta \langle U_{1+2} \rangle + \ln Z_{1+2}\right], \qquad U_{1+2} = U_1 + U_2.} \tag{3.19}$$

Independence holds when the coupling between the systems is weak. For strong coupling, the joint distribution does not factorize, and $S_{1+2} = S_1 + S_2 - I$ with the mutual information $I \geq 0$.

3.3 Energy Fluctuation, Thermodynamic Limit and Equivalence of Ensembles

Energy fluctuation and the heat capacity. The variance of the internal energy, the energy fluctuation around the controlled mean $\langle U \rangle = \bar{U}$, can be expressed through familiar thermodynamic quantities. By the chain rule,

$$\mathrm{Var}(U) = -\frac{\partial \langle U \rangle}{\partial \beta} = -\frac{\partial \langle U \rangle}{\partial T} \frac{\partial T}{\partial \beta}.$$

Since $T = 1/(k_B \beta)$, $\partial T / \partial \beta = -k_B T^2$. The isochoric heat capacity is defined as $C_V \equiv \partial \langle U \rangle / \partial T \big|_{V,N}$, at the fixed volume and particle number of the canonical ensemble. Substitution gives

$$\boxed{\mathrm{Var}(U) = k_B T^2 \, C_V.} \tag{3.20}$$

Why fluctuations vanish in the thermodynamic limit. The smallness of fluctuations is sometimes attributed to the tiny numerical value of k_B in SI units. This attribution is incorrect, because the value of k_B only reflects the choice of units. The genuine reason is that C_V and $\langle U \rangle$ are both extensive and scale linearly with the particle number N. Hence

$$\boxed{\frac{\sqrt{\mathrm{Var}(U)}}{\langle U \rangle} \sim \frac{\sqrt{N}}{N} = \frac{1}{\sqrt{N}},} \tag{3.21}$$

which vanishes as $N \to \infty$. The relative energy fluctuations of a macroscopic system are therefore negligible because of the sheer size of N. For this reason, U replaces $\langle U \rangle$ from here on, and likewise for other extensive variables. At large N, the canonical ensemble therefore behaves as the microcanonical ensemble, with negligible fluctuations and a one-to-one relation between energy and temperature.

Physical reading of the fluctuation formula. The multiplier β is smaller at higher temperature, so the entropic cost per unit of exchanged energy is smaller. The system then absorbs or releases energy more freely, consistent with the larger fluctuation $k_B T^2 C_V$.

Concavity of $S(U)$ and thermodynamic stability. For $A = U$, the concavity result and $\mathrm{Var}(U) = k_B T^2 C_V$ give

$$\boxed{\frac{\partial^2 S}{\partial U^2} = -\frac{k_B}{\mathrm{Var}(U)} = -\frac{1}{C_V\, T^2}.} \tag{3.22}$$

Concavity of $S(U)$ is therefore equivalent to $C_V > 0$, the thermodynamic stability condition, which requires that adding energy to a system raises the system temperature. This property is used again for the Legendre transformation in the next section.

3.4 Zeroth Law of Thermodynamics

Two bodies exchanging energy. Two bodies, (1) and (2), are in contact and can exchange energy. Two points need precise statements:

- The combined system $(1) + (2)$ is genuinely isolated and exchanges nothing with the outside. The total energy is a hard constraint, $U_1 + U_2 = U_{\mathrm{tot}}$, so whatever one body releases, the other receives. This constraint is a microcanonical condition on the pair as a whole.
- Each body has an entropy function $S_i(U_i, V_i, N_i)$, which is the canonical entropy derived earlier in this section. Each body carries the multiplier $\beta_i \equiv k_B^{-1}\, \partial S_i / \partial U_i$ attached to the energy of that body. The pair is microcanonical, whereas each body behaves canonically, with the other body as the energy reservoir.

Equilibrium requires equal multipliers. The two bodies are weakly coupled, so the combined entropy is additive, $S = S_1 + S_2$. Equilibrium corresponds to the maximum of S with respect to how U_{tot} is shared. The substitution $U_2 = U_{\text{tot}} - U_1$ and differentiation with respect to U_1 give

$$\frac{\partial S}{\partial U_1} = \frac{\partial S_1}{\partial U_1} + \frac{\partial S_2}{\partial U_2}\frac{\partial U_2}{\partial U_1} = \frac{\partial S_1}{\partial U_1} - \frac{\partial S_2}{\partial U_2} = 0.$$

Since $\beta_i = k_B^{-1}\,\partial S_i/\partial U_i$, this condition gives

$$\boxed{\beta_1 = \beta_2 \qquad \text{and hence} \qquad T_1 = T_2.} \tag{3.23}$$

Direction of energy flow and the zeroth law. Suppose $\beta_1 \neq \beta_2$ initially. A small energy transfer δU_1 into body (1) changes the total entropy by

$$\delta S = k_B(\beta_1 - \beta_2)\,\delta U_1.$$

The second law requires $\delta S \geq 0$, so energy flows into the body with the larger β, that is, the lower T. Energy therefore flows from high temperature to low temperature. A third body in contact with both bodies reaches the same β at equilibrium by the same argument. This transitivity is the zeroth law of thermodynamics.

Volume and particle exchange cannot be separated from energy exchange. Suppose the two bodies can also exchange volume through a movable partition, with $V_1 + V_2 = V_{\text{tot}}$ fixed. Volume exchange does not by itself imply thermal contact. A heat-insulating piston that slides freely permits volume exchange but forbids heat exchange, so $T_1 \neq T_2$ in general. The multiplier conjugate to volume is $\beta_V \equiv k_B^{-1}\,\partial S/\partial V = P/(k_B T)$. The maximum of $S = S_1(V_1) + S_2(V_2)$ over the split of V_{tot}, with U_1 and U_2 fixed, satisfies

$$\beta_{V,1} = \beta_{V,2} \qquad \Longrightarrow \qquad \frac{P_1}{T_1} = \frac{P_2}{T_2}.$$

This condition permits $P_1 \neq P_2$. However, by Newton's second law alone, a partition under a genuine pressure imbalance cannot rest. The flaw is the assumption that U_1 and U_2 stay fixed while the partition moves. A sliding wall does work $P\,dV$ on the side toward which the wall moves, so even a heat-insulating wall exchanges energy through work, $dU = -P\,dV$ for $\delta Q = 0$. If energy and volume can both be exchanged, as for a movable wall that also conducts heat, the maximum of $S_1(U_1, V_1) + S_2(U_2, V_2)$ over both splits satisfies

$$\boxed{T_1 = T_2 \quad \text{and} \quad P_1 = P_2,} \tag{3.24}$$

the physically sensible free-piston equilibrium.

Why the multipliers of V and N contain the temperature. More generally, volume or particle exchange always involves energy exchange. Each exchange contributes its own term to the first law, $-P\,dV$ or $\mu\,dN$, so the energy varies whenever either quantity varies freely. The multipliers conjugate to volume and particle number therefore always appear together with the temperature.

4 Beyond Canonical Ensemble

4.1 Internal Energy Minimization

Two ways to slice the entropy landscape. Consider the entropy $S(U, X)$ as a surface over the (X, U) plane, where X is an extensive variable that can be freely partitioned, such as a volume shared between two bodies. The level curves of S, the curves along which S is constant, resemble contour lines on a topographic map. The map can be sliced in two ways:

- Entropy maximum at fixed U. Along a horizontal line, $U = \text{const}$, the highest contour reached touches the line without crossing the line, so the contour is tangent to the line.
- Energy minimum at fixed S. Along one contour, $S = \text{const}$, the lowest point has a horizontal tangent.

Both slices therefore select a point where the level curve of S has a horizontal tangent in the (X, U) plane.

One geometric condition, two descriptions. The gradient

$$\nabla S = \left(\frac{\partial S}{\partial X}, \frac{\partial S}{\partial U} \right)$$

is perpendicular to the level curve. If $\partial S/\partial X = 0$ at a point, the gradient points purely along the U-direction, so the level curve points purely along the X-direction and is horizontal. A horizontal tangent means $dU/dX|_S = 0$, a stationary point of U along the contour. The implicit function theorem makes this condition explicit:

$$\boxed{\left.\frac{\partial U}{\partial X}\right|_S = -\frac{\partial S/\partial X\,|_U}{\partial S/\partial U\,|_X}.} \tag{4.1}$$

Since $\partial S/\partial U = 1/T$ is finite and nonzero, the left side vanishes exactly when the numerator vanishes:

$$\boxed{\left.\frac{\partial S}{\partial X}\right|_U = 0 \quad \Longleftrightarrow \quad \left.\frac{\partial U}{\partial X}\right|_S = 0.} \tag{4.2}$$

The two conditions are equivalent because they describe the same point on the same contour line.

Generality of the argument and the role of stability. This argument does not depend on entropy or energy. For any smooth function $f(x, y)$ that is monotonic in y at fixed x, the points where f is extremal in x at fixed y coincide with the points where y is extremal in x along a level set of f, because both conditions say that the contour is flat. Thermodynamics is the case $f = S$ and $y = U$, with x any extensive variable, and monotonicity follows from $\partial S/\partial U = 1/T > 0$. The thermodynamic stability condition, $\partial^2 S/\partial X^2 < 0$, adds the type of flat point, with a maximum of S at fixed U corresponding to a minimum of U at fixed S.

Which flat point: maximum of S, minimum of U. Differentiation of $\partial U/\partial X|_S = -T\,\partial S/\partial X|_U$ once more along a contour of constant S gives

$$\left.\frac{\partial^2 U}{\partial X^2}\right|_S = -T\left.\frac{d}{dX}\right|_S\left(\left.\frac{\partial S}{\partial X}\right|_U\right) - \left.\frac{\partial T}{\partial X}\right|_S\left.\frac{\partial S}{\partial X}\right|_U.$$

At the stationary point, $\partial S/\partial X|_U = 0$ and $\partial U/\partial X|_S = 0$, so the second term vanishes and the chain rule reduces the first term to $\partial^2 S/\partial X^2|_U$:

$$\boxed{\left.\frac{\partial^2 U}{\partial X^2}\right|_S = -T\left.\frac{\partial^2 S}{\partial X^2}\right|_U.} \tag{4.3}$$

Since $T > 0$ and $\partial^2 S/\partial X^2 < 0$ by stability, the right side is positive. A maximum of S at fixed U is therefore a minimum of U at fixed S.

4.2 Thermodynamic Potential

Entropy as the microcanonical thermodynamic potential. The microcanonical entropy $S(U, V, N)$ is a function of state, because the values of U, V and N determine S completely. The derivatives of S give every conjugate intensive variable, and S is the quantity extremized by the ensemble. For these reasons, S is called the thermodynamic potential of the ensemble, and (U, V, N) are the natural variables. A potential $\Psi(x_1, \ldots, x_n)$ has natural variables $(x_1, \ldots, x_n)$ when:

(i) the derivatives of Ψ with respect to the x_i give every conjugate variable, so $d\Psi$ is an exact differential in exactly these variables, with nothing missing and nothing required from outside;

(ii) the x_i are precisely the quantities externally fixed or controlled under the physical condition that Ψ describes, that is, the hard constraints of the ensemble, not an arbitrary choice of variables.

The need for a potential with natural variables (N, V, T). Because S is strictly increasing in U, the relation $S(U, V, N)$ can be inverted to give $U(S, V, N)$, which is also a function of state. The derivatives of U likewise give T, P and μ, and U is minimized at fixed (S, V, N). However, the canonical ensemble controls the temperature rather than U. The entropy is then known only as a consequence of T, not as a freely varied coordinate. The canonical ensemble therefore needs a potential that preserves the full information of the system and has the natural variables (N, V, T).

4.3 The Legendre Transformation

Constructing the transform from tangent lines. In this section, ξ denotes the slope instead of p, to avoid confusion with the probability p_i and the pressure P. Let $f(x)$ be a function. At each point x, the tangent line has slope $\xi(x) = f'(x)$ and intercept $f(x) - \xi(x)\,x$. The intercept, expressed as a function of the slope, is the Legendre transform

$$\boxed{g(\xi) = f(x) - \xi\,x, \qquad \xi = f'(x).} \tag{4.4}$$

The transform $g(\xi)$ conserves the information of $f(x)$ if f can be reconstructed from g alone.

Reconstructing f from its transform. The tangent line with slope ξ is $y = \xi x + g(\xi)$, so the tangent lines form a family with one line per ξ. At a fixed point x, the vertical gap between this family and the curve is

$$\Delta(\xi, x) = \xi x + g(\xi) - f(x).$$

A function f is convex if $-f$ is concave, that is, if the inequality in the definition of concavity is reversed. If f is convex, every tangent line lies on or below the curve, so $\Delta \leq 0$ for all ξ, with equality exactly for the line tangent at x. If f is concave, every tangent line lies on or above the curve, so $\Delta \geq 0$ with the same equality condition. In either case, the tangent line is a genuine extremum of Δ over ξ, not merely a stationary point:

$$\frac{\partial \Delta}{\partial \xi} = x + g'(\xi) = 0 \quad \Longleftrightarrow \quad x = -g'(\xi).$$

With the solution denoted $\xi^*(x)$, f is reconstructed as

$$\boxed{x = -g'(\xi), \qquad f(x) = \xi^*(x)\, x + g\big[\xi^*(x)\big].} \tag{4.5}$$

Hence f and g determine each other, and no information is lost. The transformation trades the variable x for the conjugate $\xi = f'(x)$.

4.4 Multivariable Legendre Transformation

Extending the transform to several variables. Let $f(x_1, \ldots, x_n)$ be a function of $\mathbf{x} = (x_1, \ldots, x_n)$. At each point, the tangent hyperplane has gradient $\boldsymbol{\xi}(\mathbf{x}) = \nabla f(\mathbf{x})$, with components $\xi_i = \partial f / \partial x_i$, and intercept $f - \boldsymbol{\xi} \cdot \mathbf{x}$. The Legendre transform is

$$\boxed{g(\boldsymbol{\xi}) = f(\mathbf{x}) - \boldsymbol{\xi} \cdot \mathbf{x} = f(\mathbf{x}) - \sum_{i=1}^{n} \xi_i x_i.} \tag{4.6}$$

Reconstructing f from its transform. The tangent hyperplane with gradient $\boldsymbol{\xi}$ is $y = \boldsymbol{\xi} \cdot \mathbf{x} + g(\boldsymbol{\xi})$, so the hyperplanes form a family with one member for each $\boldsymbol{\xi} \in \mathbb{R}^n$. At a fixed point $\mathbf{x}$, the vertical gap between this family and the surface is

$$\Delta(\boldsymbol{\xi}, \mathbf{x}) = \boldsymbol{\xi} \cdot \mathbf{x} + g(\boldsymbol{\xi}) - f(\mathbf{x}).$$

If f is convex, every tangent hyperplane lies on or below the surface, so $\Delta \leq 0$ for all $\boldsymbol{\xi}$, with equality exactly for the hyperplane tangent at $\mathbf{x}$. If f is concave, $\Delta \geq 0$ with the same equality condition. In either case, that is, when the Hessian of f is semidefinite everywhere, the tangent hyperplane is a genuine extremum of Δ over $\boldsymbol{\xi}$:

$$\frac{\partial \Delta}{\partial \xi_i} = x_i + \frac{\partial g}{\partial \xi_i} = 0 \quad \Longleftrightarrow \quad x_i = -\frac{\partial g}{\partial \xi_i} \qquad (i = 1, \ldots, n).$$

With the solution denoted $\boldsymbol{\xi}^*(\mathbf{x})$, f is reconstructed as

$$\boxed{x_i = -\frac{\partial g}{\partial \xi_i}, \qquad f(\mathbf{x}) = \boldsymbol{\xi}^*(\mathbf{x}) \cdot \mathbf{x} + g\big[\boldsymbol{\xi}^*(\mathbf{x})\big].} \tag{4.7}$$

No information is lost. The transformation trades the variables $x_1, \ldots, x_n$ for their conjugates $\xi_i = \partial f / \partial x_i$.

Partial Legendre transforms. In thermodynamics, a Legendre transform often acts on only a subset $I \subseteq \{1, \dots, n\}$ of the variables and leaves the rest untouched:

$$\boxed{g\big(\{\xi_i\}_{i\in I}, \{x_j\}_{j\notin I}\big) = f(\mathbf{x}) - \sum_{i\in I} \xi_i x_i, \qquad \xi_i = \left.\frac{\partial f}{\partial x_i}\right|_{\mathbf{x}}.} \tag{4.8}$$

The construction above goes through one variable at a time. Each x_i with $i \in I$ is swapped for the conjugate ξ_i, and each x_j with $j \notin I$ is held fixed as a spectator.

Convexity of partial transforms. If f is convex, a partial Legendre transform is concave in the transformed variables ξ_i and convex in the untransformed variables x_j. Since $U(S, V, N)$ is convex by thermodynamic stability, $F(T, V, N)$ is concave in T and convex in V, and $H(S, P, N)$ is concave in P and convex in S. For this reason, F is minimized over V at fixed T but not over T.

4.5 NVT Ensemble — Helmholtz Free Energy

Helmholtz free energy as the Legendre transform of U. The thermodynamic potential of the NVT ensemble, or canonical ensemble, is the Helmholtz free energy. The Helmholtz free energy is the partial Legendre transform of $U(S, V, N)$ that trades S for the conjugate $T = \partial U/\partial S|_{V,N}$:

$$F(N, V, T) = U - \left.\frac{\partial U}{\partial S}\right|_{V,N} S,$$

that is,

$$\boxed{F(N, V, T) = U - TS.} \tag{4.9}$$

Conjugate variables of F. The free energy F generates every conjugate variable. Because U depends on S, and S in turn depends on T, V and N, each derivative of F includes the contribution of S:

$$\begin{aligned}
\left.\frac{\partial F}{\partial T}\right|_{V,N} &= \frac{\partial U}{\partial S}\frac{\partial S}{\partial T} - S - T\frac{\partial S}{\partial T} = -S, \\
\left.\frac{\partial F}{\partial V}\right|_{T,N} &= \frac{\partial U}{\partial V} + \frac{\partial U}{\partial S}\frac{\partial S}{\partial V} - T\frac{\partial S}{\partial V} = -P + T\frac{\partial S}{\partial V} - T\frac{\partial S}{\partial V} = -P, \\
\left.\frac{\partial F}{\partial N}\right|_{T,V} &= \frac{\partial U}{\partial N} + \frac{\partial U}{\partial S}\frac{\partial S}{\partial N} - T\frac{\partial S}{\partial N} = \mu + T\frac{\partial S}{\partial N} - T\frac{\partial S}{\partial N} = \mu.
\end{aligned}$$

Hence

$$\boxed{\left.\frac{\partial F}{\partial T}\right|_{V,N} = -S, \qquad \left.\frac{\partial F}{\partial V}\right|_{T,N} = -P, \qquad \left.\frac{\partial F}{\partial N}\right|_{T,V} = \mu.} \tag{4.10}$$

Thus, F behaves much like U, with the same dimension and the same P and μ, whereas T has become an argument and $-S$ the conjugate.

Minimization of F from entropy maximization. The minimization of F follows from MaxEnt applied to the combined system. A system at fixed V and N exchanges energy with a reservoir at temperature T, and the pair is isolated, as in the zeroth-law setup. The reservoir gains the energy that the system loses, so $dS_{\text{res}} = -dU/T$, and the total entropy changes by

$$\boxed{dS_{\text{tot}} = dS - \frac{dU}{T} = -\frac{dF}{T} \geq 0.} \tag{4.11}$$

The maximum of the total entropy is therefore the minimum of the system free energy F. The term $-TS$ in F accounts for the reservoir.

Free energy and the partition function. Substitution of $S = k_B(\beta U + \ln Z)$ and $\beta = 1/(k_B T)$ into $F = U - TS$ gives

$$F = U - \frac{1}{k_B \beta}\, k_B\, (\beta U + \ln Z) = U - U - \frac{\ln Z}{\beta},$$

so

$$\boxed{F = -k_B T \ln Z.} \tag{4.12}$$

4.6 NPS Ensemble — Enthalpy

Enthalpy as the Legendre transform of U in the volume. For some systems, the pressure is more natural to control than the volume, because the pressure is intensive and does not depend on the system size. The thermodynamic potential is then the enthalpy, the partial Legendre transform of $U(S, V, N)$ that trades V for the conjugate $-P = \partial U/\partial V|_{S,N}$:

$$H(N, P, S) = U - \left.\frac{\partial U}{\partial V}\right|_{S,N} V,$$

that is,

$$\boxed{H(N, P, S) = U + PV.} \tag{4.13}$$

This ensemble is labeled the NPS ensemble after the natural variables of H. The enthalpy generates every conjugate variable:

$$\boxed{\left.\frac{\partial H}{\partial S}\right|_{P,N} = T, \qquad \left.\frac{\partial H}{\partial P}\right|_{S,N} = V, \qquad \left.\frac{\partial H}{\partial N}\right|_{S,P} = \mu.} \tag{4.14}$$

Enthalpy as exchanged heat, and its minimization. By the first law, at constant (N, P),

$$\boxed{\delta Q = dU + P\, dV = dH.} \tag{4.15}$$

The Clausius inequality, $\delta Q \leq T\, dS$, then gives $dH \leq T\, dS$, and at constant entropy

$$\boxed{dH \leq 0.} \tag{4.16}$$

Any spontaneous process at fixed (N, P, S) therefore proceeds toward lower H and halts at the minimum of H, which recovers the equilibrium condition established earlier. However, most processes are not isentropic, so the enthalpy is mainly used as the heat exchanged at fixed pressure. For example, the enthalpy measures the heat of a phase transition.

Labels and conventional names of ensembles. From here on, each ensemble is labeled by the natural variables of the thermodynamic potential, such as NVT for F and NPS for H. The conventional name is given alongside. The conventional label of an ensemble lists the quantities held fixed as constraints. For NPS, the conventional name is the isoenthalpic-isobaric ensemble, NPH, which fixes N, P and the enthalpy H. The NPH ensemble relates to $H(N, P, S)$ as the microcanonical ensemble NVU relates to $U(S, V, N)$, with the same state described either by the fixed energy-like quantity or by the entropy. Table 1 lists the ensembles of this section and the next.

Table 1: Natural variables, potentials and conventional names of the ensembles.

Natural variables	Potential	Conventional name
(S, V, N)	U	microcanonical (NVU)
(T, V, N)	F	canonical (NVT)
(S, P, N)	H	isoenthalpic-isobaric (NPH)
(T, V, μ)	grand potential	grand canonical (μVT)
(T, P, N)	G	isothermal-isobaric (NPT)

5 The Grand Canonical Ensemble

Adding a second constraint: particle exchange. The grand canonical ensemble adds a second constrained average, the particle number, so the system exchanges both energy and particles with the surroundings. Here, β and γ denote the multipliers of the constraints $\langle U\rangle = \bar{U}$ and $\langle N\rangle = \bar{N}$, and U_i and N_i denote the energy and particle number of microstate i. The stationarity conditions of the Lagrangian are

$$\frac{\partial \mathcal{L}}{\partial p_i} = -\ln p_i - 1 - \alpha - \beta U_i - \gamma N_i = 0, \qquad \frac{\partial \mathcal{L}}{\partial \alpha} = -\Big(\sum_i p_i - 1\Big) = 0,$$
$$\frac{\partial \mathcal{L}}{\partial \beta} = -\Big(\sum_i U_i p_i - \bar{U}\Big) = 0, \qquad \frac{\partial \mathcal{L}}{\partial \gamma} = -\Big(\sum_i N_i p_i - \bar{N}\Big) = 0.$$

The grand canonical distribution. The first condition gives $p_i = e^{-1-\alpha-\beta U_i-\gamma N_i}$. Normalization fixes α, and the remaining conditions fix the averages:

$$e^{1+\alpha} = \sum_i e^{-\beta U_i-\gamma N_i}, \qquad \bar{U} = \frac{\sum_i U_i\, e^{-\beta U_i-\gamma N_i}}{\sum_i e^{-\beta U_i-\gamma N_i}}, \qquad \bar{N} = \frac{\sum_i N_i\, e^{-\beta U_i-\gamma N_i}}{\sum_i e^{-\beta U_i-\gamma N_i}}.$$

The free parameters are now β and γ, as β alone is in the canonical ensemble, and α is a normalization factor that depends on both. The grand canonical partition function is defined as

$$\boxed{\Xi(\beta,\gamma) \equiv \sum_i e^{-\beta U_i-\gamma N_i},} \tag{5.1}$$

and the probability of a microstate with energy U_i and N_i particles is

$$\boxed{p_i = \frac{e^{-\beta U_i-\gamma N_i}}{\Xi(\beta,\gamma)}.} \tag{5.2}$$

Identifying γ with the chemical potential. The entropy of this distribution is

$$\frac{S}{k_B} = -\sum_i p_i \ln p_i = \beta\langle U\rangle + \gamma\langle N\rangle + \ln \Xi.$$

In the differential of S, with $\partial \ln \Xi/\partial\beta = -\langle U\rangle$ and $\partial \ln \Xi/\partial\gamma = -\langle N\rangle$, the terms from the implicit dependence cancel, as in the canonical case:

$$dS = k_B\left(\beta\, d\langle U\rangle + \gamma\, d\langle N\rangle\right), \qquad \text{so} \qquad \gamma = \frac{1}{k_B}\frac{\partial S}{\partial \langle N\rangle}\bigg|_{\langle U\rangle, V}.$$

The first law gives $\partial S/\partial N|_{U,V} = -\mu/T$, hence $\gamma = -\beta\mu$. Like $\beta = 1/(k_B T)$, this relation is a correspondence with classical thermodynamics. From here on, (β,μ) replaces (β,γ). The results become

$$\boxed{\Xi(\beta,\mu) = \sum_i e^{-\beta(U_i-\mu N_i)}, \qquad p_i = \frac{e^{-\beta(U_i-\mu N_i)}}{\Xi(\beta,\mu)},} \tag{5.3}$$

and the entropy is

$$\boxed{S = k_B\left[\beta\left(\langle U\rangle - \mu\langle N\rangle\right) + \ln \Xi\right].} \tag{5.4}$$

Energy and particle number are correlated. Particle exchange also exchanges energy, as the zeroth-law discussion showed. Therefore, $\langle U \rangle$ alone is expected to be a poor function of β at fixed μ. The covariance of two observables X and Y is defined as

$$\boxed{\mathrm{Cov}(X,Y) \equiv \langle XY \rangle - \langle X \rangle \langle Y \rangle, \qquad \mathrm{Var}(X) = \mathrm{Cov}(X,X).} \tag{5.5}$$

Let $w_i = U_i - \mu N_i$. For any observable X, at fixed V and μ,

$$\begin{aligned} \left.\frac{\partial \langle X \rangle}{\partial \beta}\right|_{V,\mu} &= \frac{\partial}{\partial \beta}\left(\frac{\sum_i X_i e^{-\beta w_i}}{\Xi}\right) \\ &= -\frac{\sum_i X_i w_i e^{-\beta w_i}}{\Xi} + \frac{\left(\sum_i X_i e^{-\beta w_i}\right)\left(\sum_j w_j e^{-\beta w_j}\right)}{\Xi^2} \\ &= -\langle X w \rangle + \langle X \rangle \langle w \rangle = -\mathrm{Cov}(X,w). \end{aligned}$$

For $X = U$ and $X = w$, this result gives

$$\boxed{\left.\frac{\partial \langle U \rangle}{\partial \beta}\right|_{V,\mu} = -\mathrm{Var}(U) + \mu\,\mathrm{Cov}(U,N).} \tag{5.6}$$

$$\boxed{\left.\frac{\partial}{\partial \beta}\left(\langle U \rangle - \mu \langle N \rangle\right)\right|_{V,\mu} = -\mathrm{Var}(U - \mu N).} \tag{5.7}$$

The monotonicity of $\langle U \rangle(\beta)$ depends on the sign of $\mu\,\mathrm{Cov}(U,N)$. The combination $\langle U \rangle - \mu \langle N \rangle$ always decreases with β, like $\langle U \rangle$ in the canonical ensemble. This combination follows from Ξ in the same way:

$$\boxed{\langle U \rangle - \mu \langle N \rangle = -\left.\frac{\partial \ln \Xi}{\partial \beta}\right|_{V,\mu}.} \tag{5.8}$$

Uniqueness of the multipliers. The multipliers (β, γ) are determined uniquely by the constrained means $(\langle U \rangle, \langle N \rangle)$ whenever the map between them is injective. Since $\langle U \rangle = -\partial \ln \Xi / \partial \beta$ and $\langle N \rangle = -\partial \ln \Xi / \partial \gamma$, the Jacobian of this map is minus the Hessian of $\ln \Xi$. By the covariance formula above, this Hessian is the covariance matrix of (U, N):

$$\boxed{\nabla^2_{(\beta,\gamma)} \ln \Xi = \begin{pmatrix} \mathrm{Var}(U) & \mathrm{Cov}(U,N) \\ \mathrm{Cov}(U,N) & \mathrm{Var}(N) \end{pmatrix} \equiv C.} \tag{5.9}$$

A covariance matrix is positive semidefinite. The matrix is positive definite unless some combination $aU_i + bN_i$ takes the same value in every accessible microstate. In the generic case, $\ln \Xi$ is strictly convex, so the gradient map of $\ln \Xi$ is injective. Each interior pair $(\bar{U}, \bar{N})$ therefore corresponds to exactly one pair (β, γ). This result extends the uniqueness argument of the canonical ensemble to two constraints.

5.1 Particle Number Fluctuation, Thermodynamic Limit and Equivalence of Ensembles Revisited

Particle number fluctuation. The particle number fluctuation is the variance of N. At fixed β and V, the weights depend on μ through $e^{\beta\mu N_i}$, so the derivative of the mean is

$$\left.\frac{\partial\langle N\rangle}{\partial\mu}\right|_{T,V} = \beta\left(\langle N^2\rangle - \langle N\rangle^2\right) = \beta\,\mathrm{Var}(N).$$

Hence

$$\boxed{\mathrm{Var}(N) = k_B T \left.\frac{\partial\langle N\rangle}{\partial\mu}\right|_{T,V}.} \tag{5.10}$$

The thermodynamic limit. The derivative of an extensive variable with respect to an intensive variable is extensive, so $\mathrm{Var}(N) \sim N$ and

$$\boxed{\frac{\sqrt{\mathrm{Var}(N)}}{\langle N\rangle} \sim \frac{1}{\sqrt{N}}.} \tag{5.11}$$

As for the energy in the canonical ensemble, particle fluctuations are negligible because of the sheer size of N. Therefore, N replaces $\langle N\rangle$, and U replaces $\langle U\rangle$, from here on. For very large N, the grand canonical ensemble converges to the microcanonical ensemble. The microcanonical ensemble can then be used to solve idealized versions of realistic thermodynamic systems analytically.

5.2 μVT Ensemble — Grand Potential

The grand potential. The natural variables of the grand canonical ensemble are (μ, V, T), hence the label μVT ensemble. As in the canonical ensemble, T controls the internal energy, and the chemical potential μ likewise controls the particle number. The thermodynamic potential is the grand potential, the partial Legendre transform of $U(S, V, N)$ in S and N:

$$\Phi(\mu, V, T) = U - \left.\frac{\partial U}{\partial S}\right|_{V,N} S - \left.\frac{\partial U}{\partial N}\right|_{S,V} N,$$

that is,

$$\boxed{\Phi(\mu, V, T) = U - TS - \mu N = F - \mu N.} \tag{5.12}$$

Conjugate variables of Φ. From $dU = T\,dS - P\,dV + \mu\,dN$,

$$d\Phi = dU - T\,dS - S\,dT - \mu\,dN - N\,d\mu = -S\,dT - P\,dV - N\,d\mu,$$

so

$$\boxed{\left.\frac{\partial\Phi}{\partial T}\right|_{V,\mu} = -S, \qquad \left.\frac{\partial\Phi}{\partial V}\right|_{T,\mu} = -P, \qquad \left.\frac{\partial\Phi}{\partial\mu}\right|_{T,V} = -N.} \tag{5.13}$$

Minimization of Φ at fixed (μ, V, T). At fixed V, $\delta W = 0$, so the first law gives $\delta Q = dU - \mu\, dN$. The Clausius inequality, $\delta Q \le T\, dS$, then gives $dU - \mu\, dN \le T\, dS$, and at fixed μ and T

$$\boxed{d\Phi = dU - \mu\, dN - T\, dS \le 0.} \tag{5.14}$$

Any spontaneous process at fixed (μ, V, T) therefore proceeds toward lower Φ and halts at the minimum of Φ.

Grand potential and the partition function. With $S = k_B[\beta(U - \mu N) + \ln \Xi]$ and $\beta = 1/(k_B T)$,

$$\Phi = U - \mu N - TS = (U - \mu N) - \frac{1}{k_B \beta}\, k_B\, [\beta(U - \mu N) + \ln \Xi] = -\frac{\ln \Xi}{\beta},$$

so

$$\boxed{\Phi = -k_B T \ln \Xi.} \tag{5.15}$$

5.3 NPT Ensemble — Gibbs Free Energy

The Gibbs free energy. The NPT ensemble, conventionally the isothermal-isobaric ensemble, is the ensemble most used in experiments. The ensemble controls the particle number N, the pressure P and the temperature T, and the system exchanges both energy and volume with the surroundings. Like the μVT ensemble, the NPT ensemble has two intensive natural variables, P and T. The thermodynamic potential is the Gibbs free energy, the partial Legendre transform of $U(S, V, N)$ in S and V:

$$G(T, P, N) = U - \left.\frac{\partial U}{\partial S}\right|_{V,N} S - \left.\frac{\partial U}{\partial V}\right|_{S,N} V,$$

that is,

$$\boxed{G(T, P, N) = U - TS + PV = F + PV = H - TS.} \tag{5.16}$$

Conjugate variables of G. From $dG = -S\, dT + V\, dP + \mu\, dN$,

$$\boxed{\left.\frac{\partial G}{\partial T}\right|_{P,N} = -S, \qquad \left.\frac{\partial G}{\partial P}\right|_{T,N} = V, \qquad \left.\frac{\partial G}{\partial N}\right|_{T,P} = \mu.} \tag{5.17}$$

Minimization of G at fixed (N, P, T). At fixed N, the first law gives $\delta Q = dU + P\, dV$, and the Clausius inequality gives $dU + P\, dV \le T\, dS$. At fixed P and T,

$$\boxed{dG = dU - T\, dS + P\, dV \le 0.} \tag{5.18}$$

Any spontaneous process at fixed (N, P, T) therefore proceeds toward lower G and halts at the minimum of G.

The NPT ensemble from MaxEnt. The NPT ensemble arises from the same principle as the other ensembles. The average volume is constrained, $\langle V\rangle = \bar{V}$, with a multiplier β_V, and the microstates include the volume, with the sum over V replaced by an integral if V is continuous. The general solution gives $p_i \propto e^{-\beta U_i - \beta_V V_i}$. As in the earlier cases, $\beta_V = \frac{1}{k_B}\partial S/\partial\langle V\rangle = \beta P$, because $\partial S/\partial V|_{U,N} = P/T$. With the partition function

$$\Gamma(\beta, P) = \sum_i e^{-\beta(U_i + PV_i)},$$

the entropy is $S = k_B[\beta(\langle U\rangle + P\langle V\rangle) + \ln\Gamma]$, and the potential is

$$\boxed{G = U + PV - TS = -k_B T \ln\Gamma.} \tag{5.19}$$

All five ensembles therefore follow from entropy maximization under different sets of constraints.

5.4 Euler's Homogeneous Function Theorem

Extensive variables as functions of other variables. The variables have been classified as extensive or intensive, but the dependence of one extensive variable on the others has not yet been studied. Euler's (homogeneous function) theorem is a general calculus statement for functions with a scaling property. A function $f(x_1, \ldots, x_n)$ is homogeneous of degree k if

$$\boxed{f(\lambda x_1, \ldots, \lambda x_n) = \lambda^k f(x_1, \ldots, x_n).} \tag{5.20}$$

Euler's theorem. If f is differentiable on $\mathbb{R}^n$ and homogeneous of degree k, then

$$\boxed{\sum_{i=1}^{n} x_i \frac{\partial f}{\partial x_i} = k\, f(x_1, \ldots, x_n).} \tag{5.21}$$

Proof of the theorem. The derivative of the defining relation with respect to λ, with $y_i = \lambda x_i$, follows from the chain rule:

$$\sum_{i=1}^{n} \left.\frac{\partial f}{\partial y_i}\right|_{y=\lambda x} x_i = k\lambda^{k-1} f(x_1, \ldots, x_n).$$

The theorem follows for $\lambda = 1$.

Application to the internal energy. The internal energy is homogeneous of degree one:

$$U(\lambda S, \lambda V, \lambda N) = \lambda\, U(S, V, N).$$

Euler's theorem then gives

$$\boxed{U = \left.\frac{\partial U}{\partial S}\right|_{V,N} S + \left.\frac{\partial U}{\partial V}\right|_{S,N} V + \left.\frac{\partial U}{\partial N}\right|_{S,V} N = TS - PV + \mu N.} \tag{5.22}$$

This Euler relation is a relation between values, not a statement about the variables of U. The natural variables of U remain (S, V, N), and (T, P, μ) are functions of these variables, computed from the derivatives of U. The intensive variables cannot be treated as variables of U, because scaling the system changes U but leaves (T, P, μ) unchanged.

Potentials with fewer extensive variables. The particle number N is the only extensive natural variable of the NPT ensemble, so G scales with N alone:

$$G(\lambda N, P, T) = \lambda\, G(N, P, T).$$

Euler's theorem then gives $G = \mu N$ for a single-component system, the formula commonly used in chemistry. The same argument applies to every potential, and each potential follows from subtracting the transformed terms from $U = TS - PV + \mu N$:

$$\begin{aligned} F &= U - TS = -PV + \mu N, & H &= U + PV = TS + \mu N, \\ \Phi &= U - TS - \mu N = -PV, & G &= U - TS + PV = \mu N. \end{aligned}$$

Hence

$$\boxed{\begin{aligned} F(N, V, T) &= -PV + \mu N, & H(N, P, S) &= TS + \mu N, \\ \Phi(\mu, V, T) &= -PV, & G(N, P, T) &= \mu N. \end{aligned}} \tag{5.23}$$

Ensembles with one intensive natural variable, NVT and NPS, have a potential that depends on two extensive variables. Ensembles with two intensive natural variables, μVT and NPT, have a potential that depends on one extensive variable only.

5.5 Multi-Component System

Generalization to several species. The preceding sections treated a single species, but realistic systems contain several species. Species i has the particle number N_i and the chemical potential μ_i. The pairs (μ_i, N_i) are conjugate, like the pair (μ, N) used so far, and each species adds one degree of freedom. From here on, $\sum_i \mu_i N_i$ replaces every term μN in a Legendre transformation. Euler's theorem becomes

$$\boxed{U = TS - PV + \sum_i \mu_i N_i,} \tag{5.24}$$

and all earlier results carry over.

Relations between the potentials. Figure 1 shows how Legendre transformations connect the potentials.

Summary of the ensembles. Table 2 lists the fixed variables of each ensemble and the derivatives of the potential that give the remaining variables.

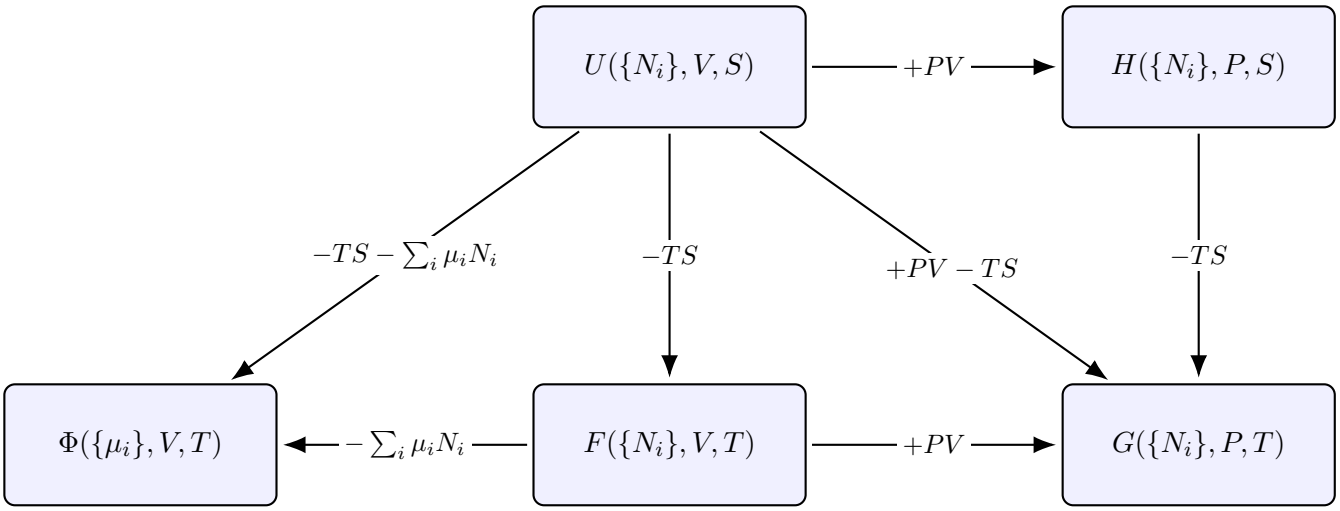


Figure 1: Legendre transformations connecting the thermodynamic potentials.

5.6 Mixing Entropy and the Gibbs Paradox

Setting and the two properties of the entropy. Let $X = (U, V, N_1, N_2)$ denote the extensive variables and $S(X) = k_B \ln \Omega(X)$ the microcanonical entropy. The derivation uses two properties, both valid in the thermodynamic limit.

- **(P1) Superadditivity.** $S(X+Y) \geq S(X)+S(Y)$. Two independent systems with variables X and Y are placed next to each other. Each pair of microstates, one from each system, is a microstate of the combined system with variables $X + Y$, and distinct pairs give distinct microstates. Hence $\Omega(X+Y) \geq \Omega(X)\Omega(Y)$, and (P1) follows from taking $k_B \ln$ of both sides.
- **(P2) Homogeneity.** $S(\lambda X) = \lambda S(X)$ for $\lambda > 0$. This property is extensivity. For identical particles, (P2) holds only if Ω counts the particles as indistinguishable, that is, with the factor $1/N!$ included.

Concavity of the entropy along the composition line. For $\lambda \in [0, 1]$, (P1) and (P2) give

$$S\big(\lambda X + (1-\lambda)Y\big) \;\geq\; S(\lambda X) + S\big((1-\lambda)Y\big) = \lambda S(X) + (1-\lambda)S(Y),$$

so S is concave in X:

$$\boxed{S\big(\lambda X + (1-\lambda)Y\big) \;\geq\; \lambda S(X) + (1-\lambda)S(Y).} \tag{5.25}$$

The total energy $2U$, the volume $2V$ and the total particle number $2N$ are now fixed. The only remaining variable is the composition, labeled by the number t of species-2 particles:

$$g(t) \equiv S(2U,\, 2V,\, 2N-t,\, t), \qquad t \in [0, 2N].$$

Table 2: Fixed variables and derivatives of the potential for each ensemble.

Ensemble	$\{N_i\}VS$	$\{N_i\}VT$	$\{N_i\}PS$	$\{\mu_i\}VT$	$\{N_i\}PT$
Conventional name	micro-canonical	canonical	isoenthalpic-isobaric	grand canonical	isothermal-isobaric
Thermodynamic potential	U	F	H	Φ	G
N_i	fixed	fixed	fixed	$-\dfrac{\partial\Phi}{\partial\mu_i}$	fixed
V	fixed	fixed	$\dfrac{\partial H}{\partial P}$	fixed	$\dfrac{\partial G}{\partial P}$
S	fixed	$-\dfrac{\partial F}{\partial T}$	fixed	$-\dfrac{\partial\Phi}{\partial T}$	$-\dfrac{\partial G}{\partial T}$
μ_i	$\dfrac{\partial U}{\partial N_i}$	$\dfrac{\partial F}{\partial N_i}$	$\dfrac{\partial H}{\partial N_i}$	fixed	$\dfrac{\partial G}{\partial N_i}$
P	$-\dfrac{\partial U}{\partial V}$	$-\dfrac{\partial F}{\partial V}$	fixed	$-\dfrac{\partial\Phi}{\partial V}$	fixed
T	$\dfrac{\partial U}{\partial S}$	fixed	$\dfrac{\partial H}{\partial S}$	fixed	fixed

The map $t \mapsto (2U, 2V, 2N - t, t)$ is affine, and a concave function composed with an affine map is concave, so g is concave. Three points on this line matter:

$$\begin{aligned} g(0) &= S(2U, 2V, 2N, 0) && \text{(all species 1)}, \\ g(2N) &= S(2U, 2V, 0, 2N) && \text{(all species 2)}, \\ g(N) &= S(2U, 2V, N, N) && \text{(half and half)}. \end{aligned}$$

Since $(N, N) = \frac{1}{2}(2N, 0) + \frac{1}{2}(0, 2N)$, the half-and-half state is the midpoint of the two pure states.

Entropy of mixing for distinguishable species. Before the partition is removed, the two systems are $X = (U, V, N, 0)$ and $Y = (U, V, 0, N)$, and the entropy of mixing is defined as

$$\boxed{\Delta S_{\text{mix}} \equiv S(2U, 2V, N, N) - S(U, V, N, 0) - S(U, V, 0, N) = g(N) - S(U, V, N, 0) - S(U, V, 0, N).} \tag{5.26}$$

Step 1: concavity at the midpoint. The concavity inequality along the line with $\lambda = \frac{1}{2}$ gives

$$g(N) \;\geq\; \tfrac{1}{2}\, g(0) + \tfrac{1}{2}\, g(2N).$$

The right side is the midpoint height of the chord that joins the two end points, so the graph of g does not lie below the chord.

Step 2: homogeneity at the end points. By (P2) with $\lambda = 2$,

$$g(0) = S(2U, 2V, 2N, 0) = 2\,S(U, V, N, 0), \qquad g(2N) = S(2U, 2V, 0, 2N) = 2\,S(U, V, 0, N).$$

With these values, the midpoint inequality becomes $g(N) \geq S(U, V, N, 0) + S(U, V, 0, N)$, that is,

$$\boxed{\Delta S_{\text{mix}} \geq 0.} \tag{5.27}$$

The same result follows directly from (P1), since $X + Y = (2U, 2V, N, N)$.

When equality holds. Equality in the midpoint inequality means that $g(N)$ lies on the chord. A concave function that touches the chord at an interior point equals the chord everywhere, because if some $g(t_0)$ were strictly above the chord, concavity would put $g(N)$ strictly above the chord too. Therefore

$$\boxed{\Delta S_{\text{mix}} = 0 \iff g \text{ is affine on } [0, 2N], \qquad \Delta S_{\text{mix}} > 0 \iff g \text{ is not affine.}} \tag{5.28}$$

In particular, $\Delta S_{\text{mix}} > 0$ whenever S is strictly concave along the composition line.

Identical particles and the paradox. For identical particles, $X = Y = (U, V, N)$, and (P2) gives

$$S(2U, 2V, 2N) = 2S(U, V, N) = S(X) + S(Y),$$

so $\Delta S_{\text{mix}} = 0$ exactly. In terms of g, if the two "species" are physically identical, S depends only on $N_1 + N_2$, so g is constant, hence affine, and the equality case applies. The difference between the two cases is whether the composition is a genuine variable of S. For distinguishable species, g is not affine, and the removal of the partition increases the entropy by the strictly positive amount ΔS_{mix}. For identical particles, g is constant and $\Delta S_{\text{mix}} = 0$. No continuity is expected between the two cases, because they differ in what the counting Ω treats as distinct. This difference is why (P2) holds for identical particles only with the $1/N!$ included.

5.7 Gibbs–Duhem Relation

The intensive variables are not independent. The Gibbs–Duhem relation states that the intensive variables $(T, P, \{\mu_i\})$ are not independent. The extensive variables are written as $x = (S, V, \{N_i\})$ and their conjugates as $\xi = (T, -P, \{\mu_i\})$, with ξ as in the Legendre transformation. Any thermodynamic potential f follows from U by exchanging the variables in a set A for their conjugates:

$$f = U - \sum_{a \in A} \xi_a x_a,$$

so that $f = U$ for $A = \varnothing$, $f = F$ for $A = \{S\}$, $f = G$ for $A = \{S, V\}$, and so on. Wherever f is differentiable in the natural variables,

$$df = \sum_{a \notin A} \xi_a\, dx_a - \sum_{a \in A} x_a\, d\xi_a.$$

By the Euler relation, $U = \sum_a \xi_a x_a$, so f is also

$$f = \sum_{a \notin A} \xi_a x_a, \qquad \text{hence} \qquad df = \sum_{a \notin A} \xi_a \, dx_a + \sum_{a \notin A} x_a \, d\xi_a.$$

The comparison of the two expressions for df gives

$$\boxed{\sum_a x_a \, d\xi_a = S \, dT - V \, dP + \sum_i N_i \, d\mu_i = 0.} \tag{5.29}$$

The result does not depend on the choice of A, so the relation holds in every ensemble. Of the n intensive variables, only $n - 1$ can be varied independently.

Why no ensemble fixes all the intensive variables. If all the extensive variables are exchanged for their conjugates, $A = \{S, V, \{N_i\}\}$, the potential becomes

$$f = U - TS + PV - \sum_i \mu_i N_i = 0.$$

This potential vanishes identically by the Euler relation. Such a potential would depend on no extensive variable at all, so nothing would fix the size of the system. A system that exchanges energy, volume and particles freely, with all of $(T, P, \{\mu_i\})$ held fixed, therefore has no ensemble of its own, and the Gibbs–Duhem relation states this fact.

Part II

Classical Thermodynamics

6 Response Functions and Stability

Scope of this section. Sections 3–5 derive the laws of thermodynamics, the potentials and the fluctuation formulas from entropy maximization. Experiments measure how one variable responds to a change of another, not the potentials themselves. This section defines these responses, relates them through exact identities and collects the conditions that stability imposes on them.

6.1 Response Functions

Measurable second derivatives. Each response function is a second derivative of a potential. The heat capacities measure the response of the entropy to the temperature, the thermal expansion coefficient measures the response of the volume to the temperature, and the compressibilities measure the response of the volume to the pressure.

$$\boxed{C_P = -T\frac{\partial^2 G}{\partial T^2}, \qquad \alpha_P = \frac{1}{V}\frac{\partial^2 G}{\partial T\,\partial P}, \qquad \kappa_T = -\frac{1}{V}\frac{\partial^2 G}{\partial P^2}.} \tag{6.1}$$

Maxwell relations. The mixed second derivatives of a potential do not depend on the order of differentiation. Each potential therefore gives one Maxwell relation.

$$\boxed{\begin{aligned} \left(\frac{\partial T}{\partial V}\right)_{S,N} &= -\left(\frac{\partial P}{\partial S}\right)_{V,N}, & \left(\frac{\partial S}{\partial V}\right)_{T,N} &= \left(\frac{\partial P}{\partial T}\right)_{V,N}, \\ \left(\frac{\partial T}{\partial P}\right)_{S,N} &= \left(\frac{\partial V}{\partial S}\right)_{P,N}, & \left(\frac{\partial S}{\partial P}\right)_{T,N} &= -\left(\frac{\partial V}{\partial T}\right)_{P,N}. \end{aligned}} \tag{6.2}$$

The cyclic relation. Three variables linked by one equation of state satisfy the cyclic relation. The slope of an isochore in the P–T plane then follows from α_P and κ_T.

$$\boxed{\left(\frac{\partial P}{\partial T}\right)_V \left(\frac{\partial T}{\partial V}\right)_P \left(\frac{\partial V}{\partial P}\right)_T = -1, \qquad \left(\frac{\partial P}{\partial T}\right)_V = \frac{\alpha_P}{\kappa_T}.} \tag{6.3}$$

Relations between the response functions. The response functions are not independent. Two exact relations connect the heat capacities and the compressibilities.

$$\boxed{C_P - C_V = \frac{TV\alpha_P^2}{\kappa_T}, \qquad \frac{C_P}{C_V} = \frac{\kappa_T}{\kappa_S}.} \tag{6.4}$$

6.2 Response Functions as Fluctuations

Heat capacity and energy fluctuation. Section 3 relates C_V to the energy fluctuation of the canonical ensemble. The relation is recalled here as the first member of a family.

$$\boxed{\mathrm{Var}(U) = k_B T^2 C_V.} \tag{6.5}$$

Compressibility and particle-number fluctuation. The fluctuation formula of the grand canonical ensemble and the Gibbs–Duhem relation give κ_T the same statistical meaning. Here, $n \equiv N/V$ denotes the number density.

$$\boxed{\frac{\mathrm{Var}(N)}{N} = n\, k_B T\, \kappa_T.} \tag{6.6}$$

6.3 Thermodynamic Stability Revisited

What Sections 3 and 4 already establish. Section 3 shows that the concavity of $S(U)$ is equivalent to $C_V > 0$. Section 4 shows that $U(S, V, N)$ is convex, so F is convex in V and G is concave in P. This subsection translates these statements into signs of the response functions and adds the conditions that Sections 3 and 4 do not state.

Positive heat capacities and compressibilities. The convexity of F in V gives $\partial^2 F/\partial V^2 = 1/(V\kappa_T) > 0$, and the fluctuation formula gives $\kappa_T > 0$ independently. The relations between the response functions then order the two heat capacities and the two compressibilities.

$$\boxed{C_P \geq C_V > 0, \qquad \kappa_T \geq \kappa_S > 0.} \tag{6.7}$$

The principle of Le Chatelier. A stable system responds to an imposed change in the direction that reduces the effect of the change. The positive response functions state this principle quantitatively.

7 The Ideal Gas

Scope of this section. The ideal gas is the first explicit model of the notes. The model consists of point particles without interactions. The canonical ensemble of Section 3 gives every property of the model, so the ideal gas is a result of the assumptions rather than an empirical law.

7.1 The Classical Partition Function

From a sum over microstates to a phase-space integral. Classical particles have continuous positions and momenta, so the sum over microstates becomes an integral over phase space. Two assumptions enter. Each microstate occupies a phase-space volume h^{3N}, and the factor $1/N!$ removes the permutations of identical particles.

$$\boxed{Z = \frac{1}{N!\,h^{3N}} \int d^{3N}q\, d^{3N}p\; e^{-\beta \mathcal{H}(q,p)}.} \tag{7.1}$$

Partition function of non-interacting particles. The Hamiltonian contains the kinetic energy only. The momentum integrals are Gaussian, and the position integrals give the volume. Here, $\lambda \equiv h/\sqrt{2\pi m k_B T}$ denotes the thermal wavelength.

$$\boxed{Z = \frac{V^N}{N!\,\lambda^{3N}}.} \tag{7.2}$$

7.2 Thermodynamics of the Ideal Gas

Equation of state, energy and heat capacity. The free energy $F = -k_B T \ln Z$ gives the pressure, and $U = -\partial \ln Z/\partial\beta$ gives the energy.

$$\boxed{PV = Nk_BT, \qquad U = \tfrac{3}{2}Nk_BT, \qquad C_V = \tfrac{3}{2}Nk_B.} \tag{7.3}$$

Entropy and chemical potential. The entropy of the ideal gas is the Sackur–Tetrode equation. The entropy is extensive only with the factor $1/N!$.

$$\boxed{S = Nk_B\left[\ln\frac{V}{N\lambda^3} + \frac{5}{2}\right], \qquad \mu = k_BT\ln\left(n\lambda^3\right).} \tag{7.4}$$

Absolute activity. The grand canonical weight $e^{\beta\mu N}$ defines the absolute activity $z \equiv e^{\beta\mu}$. For the ideal gas, the absolute activity is the number of particles in a volume λ^3.

$$\boxed{z = e^{\beta\mu} = n\lambda^3.} \tag{7.5}$$

Chemical potential and pressure. At fixed temperature, the chemical potential of the ideal gas depends on the pressure through a logarithm. This form is the reference for the fugacity of a real gas in Section 8. Here, P° denotes a reference pressure, usually 1 bar, and $\mu^\circ(T)$ the chemical potential at P°.

$$\boxed{\mu(T,P) = \mu^\circ(T) + k_B T \ln \frac{P}{P^\circ}.} \tag{7.6}$$

Response functions of the ideal gas. The general results of Section 6 take simple forms. The particle-number fluctuation equals the mean, as for a Poisson distribution.

$$\boxed{\alpha_P = \frac{1}{T}, \qquad \kappa_T = \frac{1}{P}, \qquad C_P - C_V = N k_B, \qquad \frac{\mathrm{Var}(N)}{N} = 1.} \tag{7.7}$$

Mixtures of ideal gases. In a mixture of ideal gases, each species behaves as if alone in the volume. The partial pressure $P_i = x_i P$ then replaces P in the chemical potential of species i. Here, $x_i \equiv N_i/N$ denotes the mole fraction.

$$\boxed{P = \sum_i P_i, \qquad \mu_i = \mu_i^\circ(T) + k_B T \ln \frac{x_i P}{P^\circ}.} \tag{7.8}$$

Validity of the ideal gas. The ideal gas requires two conditions. The gas must be dilute, so that the interactions are negligible, and the thermal wavelength must be small compared with the particle spacing, $n\lambda^3 \ll 1$, so that the classical limit holds.

8 Interacting Gases

Scope of this section. Real particles attract at long range and repel at short range. This section adds pair interactions to the ideal gas of Section 7, first exactly at low density and then through two closed-form models.

8.1 The Configurational Integral

Separation of the momentum integrals. Pair interactions $\phi(r)$ depend on the positions only, so the momentum integrals are those of the ideal gas. The partition function then splits into the ideal-gas factor and a configurational integral Q_N.

$$\boxed{Z = \frac{Q_N}{N!\,\lambda^{3N}}, \qquad Q_N = \int d^{3N}r \;\exp\Big[-\beta \sum_{i<j} \phi(r_{ij})\Big].} \tag{8.1}$$

8.2 The Virial Expansion

Equation of state at low density. At low density, the equation of state is a power series in $1/v$. The second virial coefficient $B_2(T)$ follows exactly from the pair potential through the Mayer function $f(r) = e^{-\beta\phi(r)} - 1$. This expansion is the rigorous link between the interactions and the deviation from ideal behavior. Here, $v \equiv V/N$ denotes the volume per particle.

$$\boxed{\frac{Pv}{k_B T} = 1 + \frac{B_2(T)}{v} + \cdots, \qquad B_2(T) = -\frac{1}{2}\int d^3 r\, f(r).} \tag{8.2}$$

Sign of the second virial coefficient. The repulsive core makes B_2 positive, and the attraction makes B_2 negative. Repulsion dominates at high temperature, and attraction dominates at low temperature.

8.3 The van der Waals Equation

Two assumptions. Two assumptions turn Q_N into a closed form. A hard core excludes the volume b per particle, and the attraction acts on each particle as a uniform mean field proportional to the density, with strength a.

$$\boxed{P = \frac{k_B T}{v - b} - \frac{a}{v^2}.} \tag{8.3}$$

Comparison with the virial expansion. The expansion of the van der Waals equation in $1/v$ gives a second virial coefficient with the same two contributions as the exact B_2.

$$\boxed{B_2^{\mathrm{vdW}}(T) = b - \frac{a}{k_B T}.} \tag{8.4}$$

Critical point and corresponding states. Below a critical temperature, each van der Waals isotherm has a loop, and the middle part of the loop has $\kappa_T < 0$, which violates the stability condition of Section 6. The loop shrinks to a horizontal inflection at the critical point, where $(\partial P/\partial v)_T = (\partial^2 P/\partial v^2)_T = 0$. The reduced variables $P_r = P/P_c$, $v_r = v/v_c$ and $T_r = T/T_c$ remove the material constants.

$$\boxed{v_c = 3b, \quad k_B T_c = \frac{8a}{27b}, \quad P_c = \frac{a}{27b^2}, \qquad \left(P_r + \frac{3}{v_r^2}\right)(3v_r - 1) = 8T_r.} \tag{8.5}$$

8.4 The Dieterici Equation

An alternative closed form. The Dieterici equation replaces the mean-field attraction by a Boltzmann factor and has the same second virial coefficient as the van der Waals equation.

$$\boxed{P = \frac{k_B T}{v - b} \exp\!\left(-\frac{a}{k_B T v}\right).} \tag{8.6}$$

The critical compressibility factor as a test. In either model, the critical compressibility factor

$$Z_c \equiv \frac{P_c v_c}{k_B T_c}$$

has one value for all substances, so Z_c tests both models against experiment (Table 3).

Table 3: Critical compressibility factor Z_c of two model equations and four fluids.

Equation or fluid	Z_c
van der Waals	$3/8 = 0.375$
Dieterici	$2/e^2 \approx 0.271$
Argon	0.291
Nitrogen	0.289
Carbon dioxide	0.274
Water	0.229

8.5 Fugacity

Definition. The fugacity f keeps the logarithmic form of the ideal-gas chemical potential of Section 7 for a real gas. The fugacity coefficient $\varphi \equiv f/P$ measures the deviation from ideal

behavior, and $\varphi \to 1$ as $P \to 0$.

$$\boxed{\mu(T,P) = \mu^\circ(T) + k_B T \ln \frac{f}{P^\circ}, \qquad \varphi \equiv \frac{f}{P}.} \tag{8.7}$$

Fugacity from the equation of state. The derivative $(\partial\mu/\partial P)_T = v$ from the Gibbs–Duhem relation, integrated from the ideal-gas limit, gives the fugacity coefficient from any equation of state.

$$\boxed{\ln\varphi = \int_0^P \left(\frac{v}{k_B T} - \frac{1}{P'}\right) dP' \approx \frac{B_2 P}{k_B T}.} \tag{8.8}$$

Fugacity and phase equilibrium. Two phases in equilibrium have equal chemical potentials, so the two phases have equal fugacities. Part III uses this equality for the vapor pressure of a real fluid.

$$\boxed{f_l(T,P) = f_g(T,P) \quad \text{at coexistence.}} \tag{8.9}$$

9 Thermodynamic Processes

Scope of this section. Sections 6–8 describe equilibrium states. This section connects the states by processes, computes the heat and the work along each process, and assembles the processes into cycles. The ideal gas of Section 7 and the van der Waals gas of Section 8 serve as the worked examples.

9.1 Quasi-Static Processes

Processes through equilibrium states. A quasi-static process passes through equilibrium states only, so the state variables are defined along the whole path. Heat and work then follow from integrals along the path, and both depend on the path.

$$\boxed{W = \int P\, dV, \qquad Q = \int T\, dS \quad \text{(reversible)}.} \tag{9.1}$$

Reversible and irreversible processes. A reversible process is quasi-static and produces no entropy, so the reverse path restores the system and the surroundings. A process with friction, heat flow across a finite temperature difference or free expansion produces entropy and is irreversible.

9.2 Elementary Processes

Isothermal, isobaric, isochoric and adiabatic processes. Each elementary process fixes one variable. The heat and the work of each process follow from the response functions of Section 6. Table 4 summarizes the four processes for an ideal gas.

Table 4: Work done by an ideal gas and heat absorbed in the four elementary processes between states 1 and 2.

Process	Fixed variable	W	Q
Isothermal	T	$Nk_BT\ln(V_2/V_1)$	W
Isobaric	P	$P(V_2-V_1)$	$C_P(T_2-T_1)$
Isochoric	V	0	$C_V(T_2-T_1)$
Adiabatic	S	$C_V(T_1-T_2)$	0

Adiabats and isotherms. An adiabat is steeper than an isotherm in the P–V plane by the factor $\gamma \equiv C_P/C_V$. For an ideal gas with constant γ, the adiabat is a power law.

$$\boxed{\left(\frac{\partial P}{\partial V}\right)_S = \gamma \left(\frac{\partial P}{\partial V}\right)_T, \qquad PV^\gamma = \text{const} \quad \text{(ideal gas)}.} \tag{9.2}$$

9.3 Heat Engines and the Carnot Cycle

Cycles, engines and efficiency. A cycle returns the system to the initial state, so $\Delta U = 0$ and the net work equals the net heat. A heat engine absorbs the heat Q_{h} from a hot reservoir, rejects the heat Q_{l} to a cold reservoir and delivers the work $W = Q_{\mathrm{h}} - Q_{\mathrm{l}}$. Here, the subscripts h and l label the hot and the cold reservoir, so that the subscript c stays reserved for the critical point.

$$\boxed{\eta \equiv \frac{W}{Q_{\mathrm{h}}} = 1 - \frac{Q_{\mathrm{l}}}{Q_{\mathrm{h}}}.} \tag{9.3}$$

The Carnot cycle. The Carnot cycle consists of two isotherms at T_{h} and T_{l} joined by two adiabats. The cycle exchanges heat with each reservoir at a single temperature, so the cycle is reversible.

$$\boxed{\frac{Q_{\mathrm{l}}}{Q_{\mathrm{h}}} = \frac{T_{\mathrm{l}}}{T_{\mathrm{h}}}, \qquad \eta_{\mathrm{C}} = 1 - \frac{T_{\mathrm{l}}}{T_{\mathrm{h}}}.} \tag{9.4}$$

The Carnot bound from the second law. The Carnot efficiency does not depend on the working substance. Over any cycle between two reservoirs, the Clausius inequality gives $Q_{\mathrm{h}}/T_{\mathrm{h}} - Q_{\mathrm{l}}/T_{\mathrm{l}} \leq 0$, so no engine exceeds the Carnot efficiency, and every reversible engine reaches the Carnot efficiency.

$$\boxed{\eta \leq \eta_{\mathrm{C}}, \qquad \oint \frac{\delta Q}{T} \leq 0.} \tag{9.5}$$

Refrigerators and heat pumps. A Carnot cycle run backward moves heat from the cold to the hot reservoir with the work W. The coefficient of performance measures the useful heat per unit work.

$$\boxed{\mathrm{COP}_{\mathrm{ref}} = \frac{Q_{\mathrm{l}}}{W} \leq \frac{T_{\mathrm{l}}}{T_{\mathrm{h}} - T_{\mathrm{l}}}, \qquad \mathrm{COP}_{\mathrm{hp}} = \frac{Q_{\mathrm{h}}}{W} \leq \frac{T_{\mathrm{h}}}{T_{\mathrm{h}} - T_{\mathrm{l}}}.} \tag{9.6}$$

Thermodynamic temperature. The ratio $Q_{\mathrm{l}}/Q_{\mathrm{h}}$ of a reversible engine depends on the two reservoir temperatures only. This ratio defines a temperature scale independent of any substance, and the scale coincides with the T of Section 3.

Other cycles. Real engines use other cycles. Each cycle combines the elementary processes and has an efficiency below the Carnot bound.

9.4 Free Expansion and Throttling

Free expansion. In a free expansion, a gas expands into vacuum inside an insulated container, so neither heat nor work is exchanged and U stays constant. The process is irreversible, and the

temperature change measures the deviation from ideal behavior.

$$\boxed{\left(\frac{\partial T}{\partial V}\right)_U = -\frac{1}{C_V}\left[T\left(\frac{\partial P}{\partial T}\right)_V - P\right].} \tag{9.7}$$

Throttling. In a throttling process, a gas flows steadily through a porous plug from high to low pressure. The enthalpy stays constant, and the Joule–Thomson coefficient gives the temperature change. The coefficient vanishes for an ideal gas.

$$\boxed{\mu_{\mathrm{JT}} \equiv \left(\frac{\partial T}{\partial P}\right)_H = \frac{V}{C_P}\left(T\alpha_P - 1\right).} \tag{9.8}$$

Joule–Thomson inversion. The attraction cools a throttled gas, and the hard core heats the gas. At low density, the van der Waals gas of Section 8 cools on throttling below the inversion temperature. Gas liquefaction relies on this cooling.

$$\boxed{k_B T_{\mathrm{inv}} = \frac{2a}{b} = \frac{27}{4}\,k_B T_c.} \tag{9.9}$$

Part III

Phase Transitions

10 First-Order Phase Transitions and Phase Diagrams

Phases and phase transitions. A phase of a system is a region of the space of natural variables in which the thermodynamic potential is analytic, that is, infinitely differentiable, with no kinks or divergences. A phase transition is a point where the potential fails to be analytic. Strictly, such singularities appear only in the thermodynamic limit $N \to \infty$, because the partition function of a finite system is analytic.

The Gibbs free energy and the composition variables. Phases are usually studied at fixed particle numbers, pressure and temperature for two reasons. Experimentally, these variables are easy to control. Structurally, the phases inside such a system exchange energy, volume and particles with each other, so $(\{\mu_i\}, P, T)$ are equal in all phases at equilibrium. The natural choice of potential is then the Gibbs free energy $G(T, P, \{N_i\})$. Because G is extensive,

$$G(T, P, \{N_i\}) = N\, g(T, P, \{x_i\}), \qquad x_i = N_i/N,$$

so the total size N does not affect the analyticity of G, and only g matters. Of the C mole fractions $\{x_i\}$ of a C-component system, only $C-1$ are independent, because $\sum_i x_i = 1$. A phase is therefore a region of the intensive variables $(T, P, \{x_i\})$, which span a $(C-1)+2 = (C+1)$-dimensional space. The collection of all phases in this space is the phase diagram.

First-order transitions as creases of g. At equilibrium, g is continuous, so a phase transition is never a jump in the potential itself. With g as one more axis on the $(C+1)$-dimensional space $(T, P, \{x_i\})$, the potential forms a hypersurface $g(T, P, \{x_i\})$. A first-order transition is a crease (kink) of this hypersurface, across which g is continuous but the first derivatives of g jump:

$$s = -\frac{\partial g}{\partial T}, \qquad v = \frac{\partial g}{\partial P},$$

which are the entropy and the volume per particle. Equivalently, $S = -\partial G/\partial T$ and $V = \partial G/\partial P$ are discontinuous, which gives a latent heat $L = T\Delta S$ and a volume change ΔV. Melting and boiling are examples. On the transition locus, the coexisting phases share the same $(\{\mu_i\}, P, T)$ but differ in s, v and composition. For this reason, the transition appears as the boundary of a coexistence region in the phase diagram.

Higher-order transitions. Higher-order transitions also exist. The order of a transition is the order of the lowest derivative of g that is discontinuous or singular. For higher-order transitions, the hypersurface has no kink, and the singularity appears in the curvature. These transitions include the continuous transitions, such as the ferromagnetic Curie point, where response functions such as the heat capacity diverge. A later section treats these transitions.

10.1 Gibbs Phase Rule

Counting the degrees of freedom. Suppose $\mathcal{P}$ phases of a system with C components coexist. At equilibrium, all the phases share the same $C+2$ intensive variables $(T, P, \mu_1, \ldots, \mu_C)$. Each

phase has its own Gibbs–Duhem relation,

$$S_\alpha\, dT - V_\alpha\, dP + \sum_i N_{i,\alpha}\, d\mu_i = 0, \qquad \alpha = 1, \ldots, \mathcal{P},$$

which imposes $\mathcal{P}$ independent constraints on these variables. The number of degrees of freedom F is the number of variables minus the number of constraints,

$$\boxed{F + \mathcal{P} = C + 2.} \tag{10.1}$$

This relation is the Gibbs phase rule. The rule counts only intensive degrees of freedom, so the amount of each phase remains free and is not counted.

Limits of the phase rule. The rule in this form applies only if $(T, P, \mu_1, \ldots, \mu_C)$ are the only intensive variables. Magnetic systems, for example, also depend on the magnetic field $\vec{B}$, which adds another intensive variable to the count.

10.2 Clapeyron Equation and Fixed-Composition (P–T) Phase Diagram

The two-phase coexistence curve. Temperature and pressure appear in every phase diagram, whatever the number of components. The P–T plane at fixed composition $\{x_i\}$ is a two-dimensional slice of the $(C+1)$-dimensional phase diagram. Only P and T vary in this slice, so the slice has two degrees of freedom. Two phases, 1 and 2, coexist when every chemical potential matches,

$$\mu_i^{(1)}(T, P) = \mu_i^{(2)}(T, P), \qquad i = 1, \ldots, C.$$

These conditions relate P and T and reduce the degrees of freedom from two to one. The coexistence locus is therefore a curve in the plane.

The Clapeyron equation. Along this curve, the two phases keep equal chemical potentials, so $d\mu_i^{(1)} = d\mu_i^{(2)}$, and the Gibbs–Duhem relation at fixed composition gives

$$-s_1\, dT + v_1\, dP = -s_2\, dT + v_2\, dP,$$

which is the Clapeyron equation

$$\boxed{\frac{dP}{dT} = \frac{\Delta s}{\Delta v}}, \qquad \Delta s = s_2 - s_1, \quad \Delta v = v_2 - v_1. \tag{10.2}$$

At coexistence, the transition proceeds reversibly at fixed T and P, so the latent heat per particle is $\ell = T\Delta s$, and

$$\boxed{\frac{dP}{dT} = \frac{\ell}{T\,\Delta v}.} \tag{10.3}$$

Three-phase coexistence and the triple point. If a third phase, 3, also coexists, then

$$\mu_i^{(1)}(T,P) = \mu_i^{(2)}(T,P) = \mu_i^{(3)}(T,P), \qquad i = 1, \ldots, C.$$

Each pair of phases has its own coexistence curve, $\mu^{(1)} = \mu^{(2)}$, $\mu^{(2)} = \mu^{(3)}$ and $\mu^{(1)} = \mu^{(3)}$. A point on the first two curves satisfies $\mu^{(1)} = \mu^{(2)} = \mu^{(3)}$, so the point lies on the third curve as well, and only two of the three conditions are independent. Two independent equations in the two unknowns (T, P) leave no degrees of freedom, so three phases can coexist only at an isolated point, the triple point, where the three two-phase curves, called triple curves, intersect.

10.3 Unary System

Definition and phase rule. A unary system has only one component, $C = 1$, and is a special case of the fixed-composition slice with $x = 1$. The unary system is the building block of higher-dimensional phase diagrams and introduces the terminology used for them. For $C = 1$, the phase rule reads $F = 3 - \mathcal{P}$, so the number of coexisting phases $\mathcal{P}$ fixes the number of free variables.

Bivariant, univariant and invariant equilibria. Within a single phase ($\mathcal{P} = 1$, $F = 2$), P and T vary freely, and the system is in bivariant equilibrium. When two phases coexist ($\mathcal{P} = 2$, $F = 1$), only one of P and T varies freely, and the coexistence curve $P(T)$, with the slope given by the Clapeyron equation, fixes the other. The system is then in univariant equilibrium, and the two-phase coexistence locus is a univariant curve. When three phases coexist ($\mathcal{P} = 3$, $F = 0$), no variable varies freely. The system is in invariant equilibrium, and the point of three-phase coexistence is an invariant point, the triple point.

The chemical potential of each phase. For $C = 1$, the Euler relation gives $G = \mu N$, so the Gibbs free energy per particle is the chemical potential, $g = \mu$. Each phase α has a separate function $\mu_\alpha(T, P)$, defined wherever the phase exists at least as a local minimum of G. The Gibbs–Duhem relation per particle, $d\mu = -s\,dT + v\,dP$, gives the slopes

$$\boxed{\left.\frac{\partial \mu_\alpha}{\partial T}\right|_P = -s_\alpha < 0, \qquad \left.\frac{\partial \mu_\alpha}{\partial P}\right|_T = v_\alpha > 0.} \tag{10.4}$$

At fixed P, every curve $\mu_\alpha(T)$ therefore decreases with temperature, and the curve of the phase with the larger entropy decreases faster. With the heat capacity per particle at constant pressure, $c_P \equiv T\,\partial s/\partial T|_P > 0$, the curvature $\partial^2\mu_\alpha/\partial T^2|_P = -c_P/T$ is negative, so each curve is also concave. In most substances, $s_{\text{vapor}} > s_{\text{liquid}} > s_{\text{solid}}$, so the solid has the lowest μ at low temperature, the liquid at intermediate temperature and the vapor at high temperature (Fig. 2).

The stable phase minimizes the chemical potential. The Gibbs free energy $G = N\mu$ is minimized at fixed (N, P, T). The stable phase is therefore the phase with the lowest chemical potential, and the equilibrium chemical potential is the lower envelope of the single-phase curves:

$$\boxed{\mu(T,P) = \min_\alpha \mu_\alpha(T,P).} \tag{10.5}$$

Two curves cross at a transition temperature, such as the melting temperature T_m or the boiling temperature T_b. At the crossing, the envelope is continuous, but the slope jumps from $-s_1$ to $-s_2$. This kink is the crease of g in a one-dimensional slice, and the jump in slope gives the latent heat per particle, $\ell = T\Delta s$.

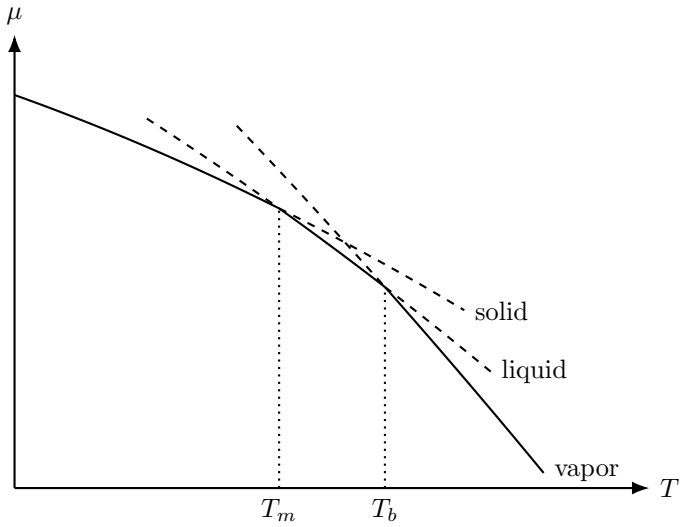


Figure 2: Chemical potentials of the solid, liquid and vapor phases at fixed pressure. Solid lines: stable branches (lower envelope). Dashed lines: metastable branches.

Metastable phases and polymorphs. Beyond a crossing, the curve of the former phase continues above the envelope (dashed lines in Fig. 2). A phase on such a branch is metastable, because G is at a local but not a global minimum. The phase persists until a fluctuation nucleates the stable phase. A supercooled liquid below T_m and a superheated liquid above T_b are examples. A unary system may also have several solid phases with different crystal structures, called polymorphs, each with a separate curve $\mu_\alpha(T, P)$. For example, the black γ phase of $CsPbI_3$ is a metastable polymorph at room temperature, with the yellow δ phase as the stable phase.

Metastable extensions of the triple curves and polymorphs. Each triple curve continues beyond the triple point as a metastable equilibrium. On the extension of the curve $\mu^{(1)} = \mu^{(2)}$, the difference $\mu^{(3)} - \mu^{(1)}$ has changed sign at the triple point, so $\mu^{(3)}$ is the lowest of the three. The extension of each triple curve therefore lies inside the field of the phase that does not take part in that curve. Equivalently, two neighboring triple curves never enclose an angle larger than 180° at the triple point. For example, an undercooled liquid below the triple point coexists with its vapor along the extension of the liquid–vapor curve. Four phases cannot coexist in a unary system, so the boundary of each additional polymorph meets the existing boundaries at new triple points, such as points where two solids and the liquid or two solids and the vapor coexist (Fig. 3). In Fig. 3(b), the metastable extensions of the α–vapor and liquid–vapor curves cross inside the β field at a metastable triple point $\alpha + L + V$. From this point, the metastable melting curve of α runs through the β field to the stable triple point $\alpha + \beta + L$ and continues above that point as the stable melting curve of α. Sulfur, with rhombic α and monoclinic β, has this diagram.

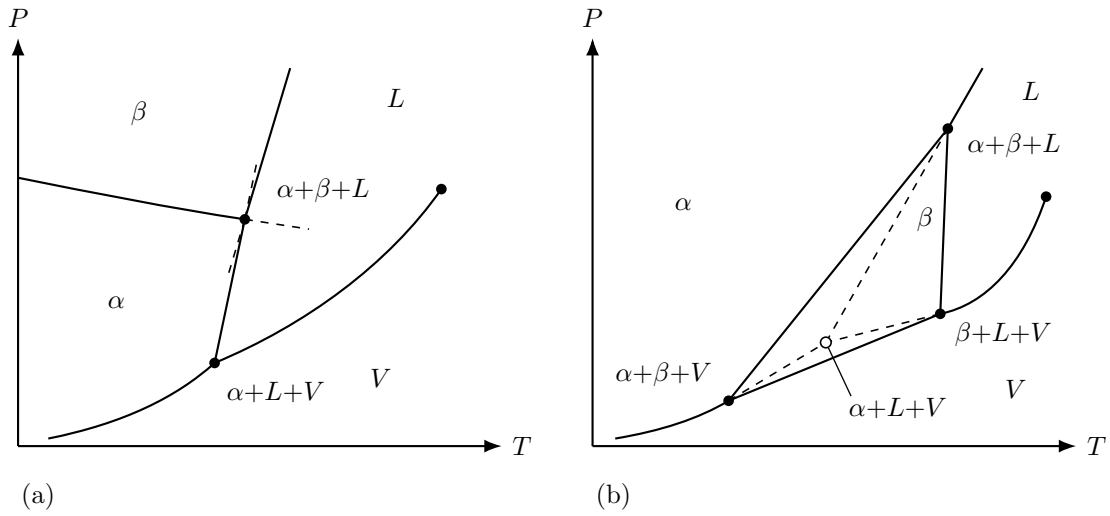


Figure 3: Schematic P–T diagrams of a unary system with two solid polymorphs α and β. (a) Triple point $\alpha+\beta+L$ on the melting curve. (b) Triple point $\alpha+\beta+V$ on the sublimation curve, with a closed β field. Dashed lines: metastable extensions. Open circle: metastable triple point.

Slopes of the coexistence curves. The Clapeyron equation fixes the slope of each univariant curve in the P–T diagram (Fig. 4). Melting involves a small volume change, so the solid–liquid curve is steep. For most substances, $\Delta v > 0$ on melting, and the slope is positive. Water is an exception, because ice is less dense than liquid water, so $\Delta v < 0$ and the melting curve has a negative slope. On the liquid–vapor and solid–vapor curves, the vapor volume dominates, $\Delta v \approx v_{\text{vapor}}$. Both ℓ and Δv are positive there, so the vapor pressure rises with temperature. At the triple point, the entropy changes around a closed cycle of the three phases sum to zero, so the latent heats satisfy $\ell_{\text{sub}} = \ell_{\text{fus}} + \ell_{\text{vap}}$. Since $\ell_{\text{sub}} > \ell_{\text{vap}}$ and Δv is nearly the same for both, the solid–vapor curve is steeper than the liquid–vapor curve at the triple point.

Thermal expansion and compressibility. The volume per particle of one phase is a function $v(T,P)$, so a change of state changes v by

$$dv = \left(\frac{\partial v}{\partial T}\right)_P dT + \left(\frac{\partial v}{\partial P}\right)_T dP. \tag{10.6}$$

The two partial derivatives, divided by v, define the thermal expansion coefficient α_P and the isothermal compressibility κ_T,

$$\boxed{\alpha_P \equiv \frac{1}{v}\left(\frac{\partial v}{\partial T}\right)_P, \qquad \kappa_T \equiv -\frac{1}{v}\left(\frac{\partial v}{\partial P}\right)_T.} \tag{10.7}$$

The subscript P distinguishes α_P from the phase label α. The minus sign makes κ_T positive, because stability requires the volume to decrease when the pressure rises. A liquid is a condensed phase. The particles of a liquid are already packed at contact distance, and the strong short-range repulsion between them resists any further decrease of the volume. A rise in pressure therefore barely changes the volume of a liquid, so $(\partial v/\partial P)_T$ and κ_T of a liquid are tiny. A vapor, with

particles far apart, compresses easily. For water at 100 °C, the liquid has $\alpha_P \approx 7.5 \times 10^{-4}$ K^{-1} and $\kappa_T \approx 4.9 \times 10^{-10}$ Pa^{-1}, while the vapor at the same temperature and 1 atm has $\alpha_P \approx 3 \times 10^{-3}$ K^{-1} and $\kappa_T \approx 1 \times 10^{-5}$ Pa^{-1}. The vapor is thus about four times more expandable and about 2×10^4 times more compressible than the liquid.

The critical point. The liquid–vapor curve does not continue indefinitely. Along the curve, T and P change together, $dP = (dP_{\text{sat}}/dT)\,dT$, so the volume of each coexisting phase changes as

$$\boxed{\frac{1}{v}\frac{dv}{dT}\bigg|_{\text{coex}} = \alpha_P - \kappa_T \frac{dP_{\text{sat}}}{dT}.} \tag{10.8}$$

Heating expands each phase through α_P, and the rising saturation pressure compresses each phase through κ_T. The cyclic relation among the partial derivatives of $v(T, P)$ gives the slope of an isochore, a line of constant v, in the P–T plane,

$$\left(\frac{\partial P}{\partial T}\right)_v = -\frac{(\partial v/\partial T)_P}{(\partial v/\partial P)_T} = \frac{\alpha_P}{\kappa_T}. \tag{10.9}$$

Hence

$$\boxed{\frac{1}{v}\frac{dv}{dT}\bigg|_{\text{coex}} = \kappa_T \left[\left(\frac{\partial P}{\partial T}\right)_v - \frac{dP_{\text{sat}}}{dT}\right].} \tag{10.10}$$

Since $\kappa_T > 0$, a phase becomes denser along the curve when the coexistence curve is steeper than the isochore of that phase, and less dense when the curve is flatter. The liquid barely compresses, so the isochores of the liquid are steep. For water at 100 °C, $\alpha_P/\kappa_T \approx 1.5$ MPa/K for the liquid, while $dP_{\text{sat}}/dT \approx 3.6$ kPa/K. The vapor compresses easily, so the isochores of the vapor are flat, with $\alpha_P/\kappa_T \approx 0.3$ kPa/K. The coexistence curve thus lies between the two isochore slopes. The liquid expands and the vapor becomes denser along the curve. Steam tables confirm the result. Between 100 °C and 105 °C, v_l rises by about 0.08% per kelvin, and v_g falls by about 3% per kelvin. The volume difference $\Delta v = v_g - v_l$ therefore shrinks from both sides as T rises.

Thermodynamics alone does not fix where this trend ends, but experiment and models such as the van der Waals equation show that Δv keeps shrinking and reaches zero at a finite temperature. The Clapeyron equation, $\Delta s = \Delta v\, dP/dT$, then gives $\Delta s = 0$ and $\ell = 0$. The two coexisting phases have the same v, the same s and the same μ, so the two phases become identical, and no coexistence curve can continue beyond this critical point (T_c, P_c). The flat two-phase segment of the isotherms in the P–v plane (Fig. 5) shrinks to a single point at the critical volume v_c, so the critical isotherm has $(\partial P/\partial v)_T = 0$ at v_c. The compressibility, $\kappa_T = -1/[v(\partial P/\partial v)_T]$, therefore diverges at the critical point. Above T_c, liquid and vapor can no longer be distinguished, and the fluid is called supercritical. A path that circles around the critical point connects the liquid to the vapor without crossing any coexistence curve, so the potential stays analytic along the whole path. By the definition of a phase, liquid and vapor are therefore one fluid phase, and the distinction between them exists only below T_c. No such path exists between solid and liquid, because the two phases differ in symmetry, and a symmetry cannot change continuously. The solid–liquid curve therefore has no critical point. The critical point itself is a continuous transition and is treated later.

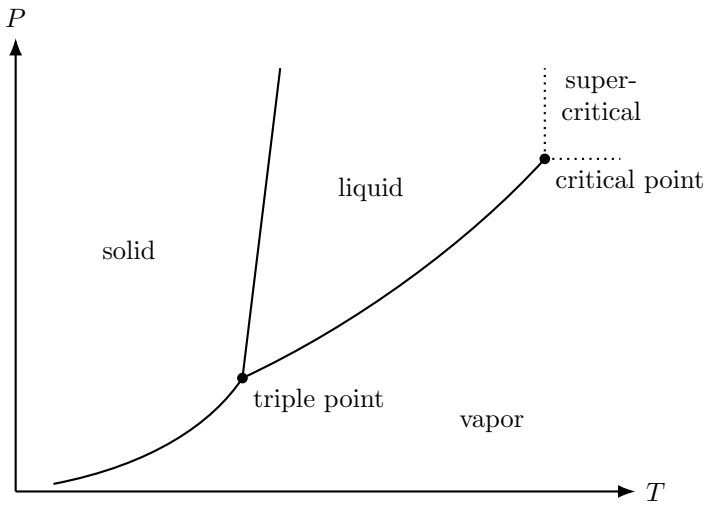


Figure 4: Schematic P–T diagram of a unary system with a triple point and a liquid–vapor critical point.

Two-phase coexistence in the P–v plane. From here on, $v \equiv V/N$ and $f \equiv F/N$ denote the volume and the Helmholtz free energy per particle, $n_\alpha \equiv N_\alpha/N$ denotes the fraction of particles in phase α, and the subscripts l and g label the liquid and the vapor (gas). The P–T diagram does not show the amount of each phase. An isotherm in the P–v plane below T_c does (Fig. 5). As the vapor is compressed, the pressure rises until the vapor pressure $P_{\text{sat}}(T)$ is reached at $v = v_g$. Further compression converts vapor into liquid at constant pressure until $v = v_l$, and only then is the liquid compressed. Between v_l and v_g, the system is a mixture of liquid and vapor at the same (T, P_{sat}). The phase rule allows exactly this mixture, because coexistence fixes P at fixed T but leaves the amounts of the two phases free.

The lever rule. Particle number and volume are conserved:

$$n_l + n_g = 1, \qquad n_l v_l + n_g v_g = v.$$

These two equations give the lever rule

$$\boxed{n_l = \frac{v_g - v}{v_g - v_l}, \qquad n_g = \frac{v - v_l}{v_g - v_l}.} \tag{10.11}$$

Equivalently, $n_l\,(v - v_l) = n_g\,(v_g - v)$. The overall state v is the fulcrum of a lever with arms $v - v_l$ and $v_g - v$, and each phase fraction is proportional to the arm on the opposite side. The overall state therefore lies closer to the dominant phase. The rule holds for any conserved extensive quantity per particle, and the binary system applies the same rule to the composition.

The common tangent construction. The free energy per particle at fixed T gives the coexisting volumes v_l and v_g. A phase at (T, v) has the pressure $P = -\partial f/\partial v$ and, by the Euler relation $F = -PV + \mu N$, the chemical potential $\mu = f + Pv$. The tangent line to f at v therefore has slope $-P$ and intercept μ at $v = 0$, which is the Legendre construction of Section 4 again. Coexistence

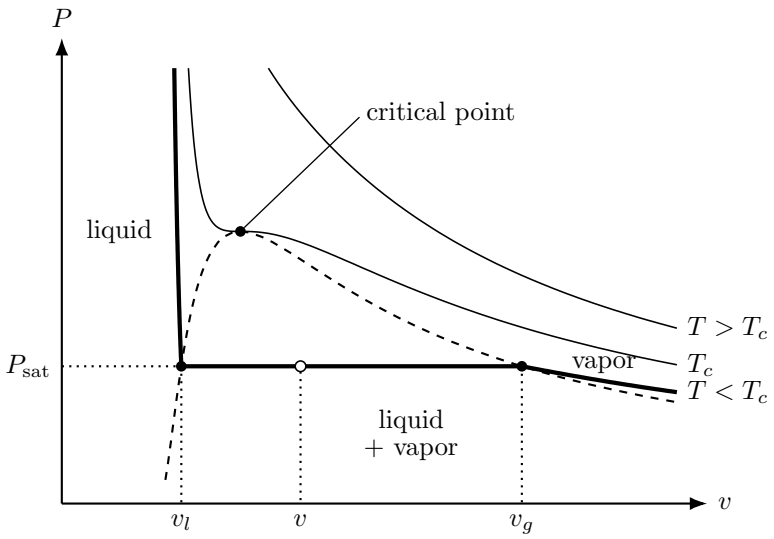


Figure 5: Isotherms in the P–v plane, computed from the van der Waals equation with the coexisting volumes from the common tangent. Thick line: isotherm below T_c, with the two-phase segment at P_{sat} between v_l and v_g. Open circle: overall state v of a liquid–vapor mixture. Dashed curve: boundary of the two-phase region, with the critical point at the top.

requires equal pressures and equal chemical potentials. Equal pressures give equal slopes, and equal chemical potentials give $f_g(v_g) - f_l(v_l) = -P_{\text{sat}}(v_g - v_l)$, so the chord between the two points has the same slope as the tangents:

$$\boxed{f_l'(v_l) = f_g'(v_g) = \frac{f_g(v_g) - f_l(v_l)}{v_g - v_l} = -P_{\text{sat}}.} \tag{10.12}$$

The two tangent lines therefore coincide in a common tangent (Fig. 6).

The mixture lies on the tangent and minimizes F. For $v_l < v < v_g$, the mixture has the free energy per particle

$$f_{\text{mix}}(v) = n_l\, f_l(v_l) + n_g\, f_g(v_g).$$

By the lever rule, n_l and n_g are linear in v, so f_{mix} is the straight segment between the two tangent points, that is, the common tangent. Where the single-phase curve lies above this segment, the mixture has the lower free energy, and the minimization of F at fixed (N, V, T) selects the mixture. The equilibrium $f(v)$ is therefore the convex hull of the single-phase curves, the largest convex function that lies below all of them. This construction restores the convexity of F in V that stability requires and explains why the pressure, $-\partial f/\partial v$, stays constant across the two-phase region.

Maxwell's equal-area rule. A model equation of state, such as that of van der Waals, may give a single isotherm $P(v)$ with a non-monotonic loop below T_c. The common tangent then becomes a condition on $P(v)$. The integral of $P = -\partial f/\partial v$ along the model isotherm gives

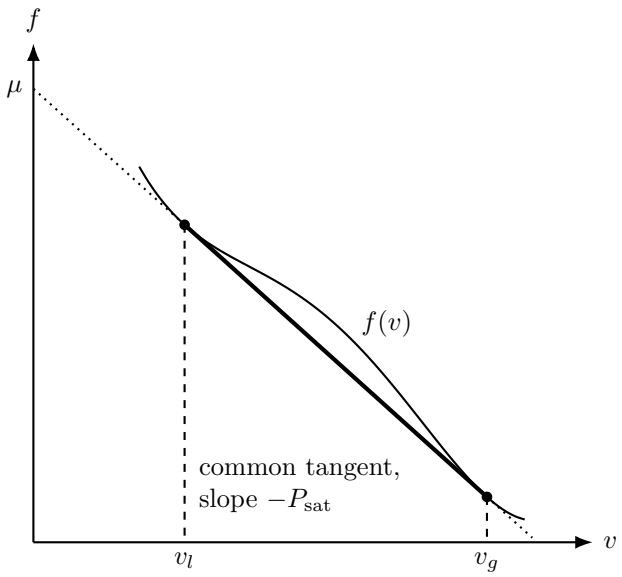


Figure 6: Common tangent construction on the free energy per particle at fixed $T < T_c$. Thick line: equilibrium two-phase mixture (convex hull).

$f(v_l) - f(v_g) = \int_{v_l}^{v_g} P\,dv$, and the common tangent gives $f(v_l) - f(v_g) = P_{\text{sat}}(v_g - v_l)$. Hence

$$\boxed{\int_{v_l}^{v_g} P(v)\,dv = P_{\text{sat}}\,(v_g - v_l).} \tag{10.13}$$

The horizontal line $P = P_{\text{sat}}$ therefore cuts the loop into two regions of equal area.

Metastable and unstable uniform states. Between v_l and v_g, the single-phase curve is a bookkeeping curve. The curve gives the free energy of a system forced to stay uniform at the density $1/v$, and the comparison with the common tangent shows that splitting into two phases lowers F. The uniform state is therefore never the equilibrium state in this range, and below T_c the equilibrium transition from liquid to vapor always passes through a two-phase mixture. The inflection points of f, where $\partial^2 f/\partial v^2 = -\partial P/\partial v = 0$, are the spinodal points. On a van der Waals isotherm, the spinodal points are the minimum and the maximum of the loop. Between v_l and the first spinodal point, and between the second spinodal point and v_g, f is convex. A small density fluctuation raises F, so the uniform phase is metastable. The mixture forms only after a large fluctuation creates a nucleus of the other phase, because the interface of the nucleus costs free energy. Superheated water in a smooth container and supercooled vapor in a cloud chamber are examples. Between the two spinodal points, f is concave. A small fluctuation splits the region into a denser and a thinner part with an average free energy below the curve, so the uniform state is unstable and separates at once by spinodal decomposition. Only a model that enforces uniformity, such as the van der Waals equation, produces a single-phase curve with such a stretch. The exact free energy of a large system already includes the mixed states and equals the convex hull. A continuous path from liquid to vapor without a two-phase mixture exists only around the critical point.

Metastable coexistence in the presence of a third phase. A common tangent never lies above the two single-phase curves that the tangent touches, because the convex hull selects the lowest line. A chord between two points on a convex stretch of one curve lies above that curve, so a stable single phase does not split. With a third phase, however, the common tangent of two phases may lie above the curve of the third phase (Fig. 7). At the triple point, a single line touches the curves of the solid, the liquid and the vapor, so the three phases have the same P and the same μ. Below the triple point, the solid curve drops below this line. The solid–vapor tangent then forms the convex hull, and the liquid–vapor tangent lies above the solid curve. Liquid and vapor still have equal pressures and equal chemical potentials on this tangent. The tangent to the solid curve with the same slope, however, has a lower intercept, so the solid has a lower chemical potential at the same pressure. The liquid–vapor equilibrium is therefore metastable. This equilibrium is the supercooled liquid with the vapor on the extension of the liquid–vapor curve below the triple point. A metastable two-phase equilibrium is thus a common tangent that lies above the curve of another phase.

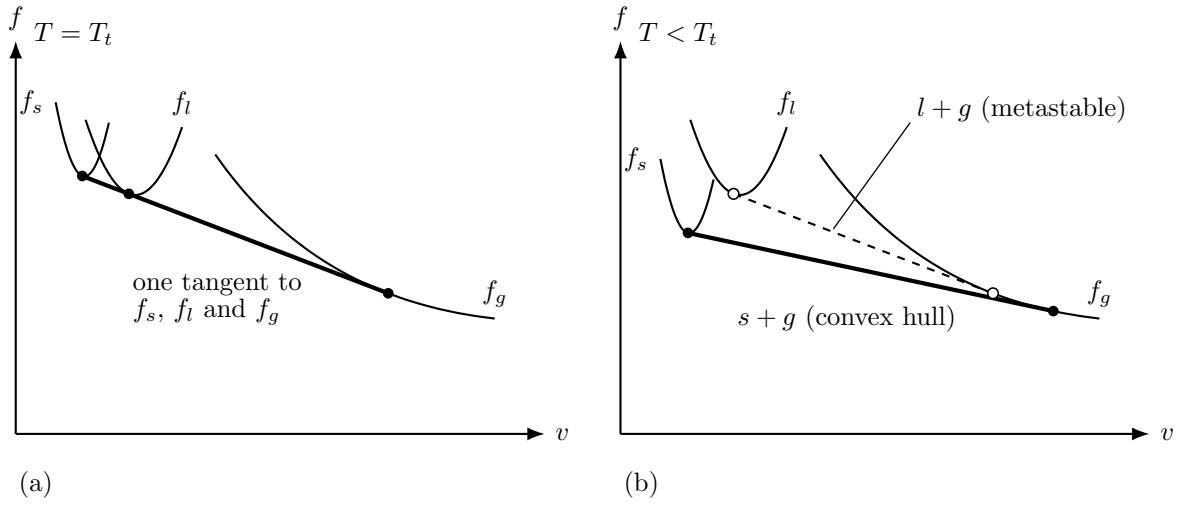


Figure 7: Free energy per particle of the solid (f_s), the liquid (f_l) and the vapor (f_g) at fixed T. (a) Triple point T_t, with one tangent to all three curves. (b) Below the triple point. Thick line: solid–vapor tangent (convex hull). Dashed line: liquid–vapor tangent above the solid curve (metastable coexistence).

11 Binary System

11.1 The X–P–T Diagram

Composition variable and phase rule. A binary system has two components, $C = 2$, so one independent variable, $X = x_2$ with $x_1 = 1 - X$, describes the composition. The phase diagram is therefore three-dimensional in (P, T, X). The diagram can be pictured as a stack of fixed-composition P–T diagrams, one for each value of X. The phase rule reads $F = 4 - \mathcal{P}$, so each unary structure gains one dimension:

- Tervariant volume ($\mathcal{P} = 1$, $F = 3$). The bivariant regions of the unary diagram become volumes in which P, T and X all vary freely without a phase change. The cross section of each volume changes with X.
- Bivariant surface ($\mathcal{P} = 2$, $F = 2$). The univariant curves become surfaces. Two coexisting phases generally have different compositions, so each phase has its own bivariant surface. The segment that joins the compositions of the two coexisting phases at one (P, T) is a tie line, and the tie lines connect the two surfaces. The unary system has no tie lines, because coexisting phases of one component have the same composition. The region between the surfaces is the two-phase region, in which the overall composition fixes the amount of each phase.
- Univariant equilibrium ($\mathcal{P} = 3$, $F = 1$). The invariant points become univariant equilibria, along which only one variable varies freely. At each (P, T) of such an equilibrium, the three phases have three different compositions. All three compositions share the same P and T, so the three points lie on one isobaric and isothermal line, the three-phase tie line. As (P, T) moves along the equilibrium, the three-phase tie lines sweep a ruled surface in the X–P–T space.
- Invariant equilibrium ($\mathcal{P} = 4$, $F = 0$). A new kind of invariant equilibrium appears, at which none of P, T or the four phase compositions varies. The four phases lie on a single isobaric and isothermal four-phase tie line. Every subset of the four phases coexists nearby, so $\binom{4}{3} = 4$ three-phase, $\binom{4}{2} = 6$ two-phase and $\binom{4}{1} = 4$ one-phase equilibria meet at this equilibrium, which is called the quadruple point.

The P–T projection. Every two-phase, three-phase and four-phase tie line is isobaric and isothermal, so each one projects to a single point of the P–T plane. The two-phase curves of the pure components lie on the faces $X = 0$ and $X = 1$ and project unchanged. A three-phase ruled surface projects to a curve, and a four-phase equilibrium projects to a point. One-phase and two-phase regions overlap in the projection, so the areas of a P–T projection cannot be labeled uniquely, and only the curves and points carry a unique meaning. The projection applies to any number of components and is used in Section 11.13.

11.2 Gibbs Free Energy of a Binary Solution

The Gibbs free energy as a sum over the components. From here on, $\mu_i^{\circ,\alpha}(T,P)$ denotes the chemical potential of pure component i in the structure of phase α, and $g^\alpha(T,P,X) \equiv G^\alpha/N$ denotes the Gibbs free energy per particle of phase α, with $N = N_1 + N_2$. The superscript α is dropped when only one phase is discussed. The composition dependence of g decides which phases coexist in a binary system. By the Euler relation of the multi-component system, the Gibbs free energy of a phase is the sum of the contributions of the two components:

$$\boxed{G = N_1\mu_1 + N_2\mu_2, \qquad g(T,P,X) = (1-X)\,\mu_1 + X\,\mu_2,} \tag{11.1}$$

where each μ_i depends on (T,P,X). This relation holds for any solution and requires no model of the interactions. The tangent intercepts, the common tangent and the lever rule below follow from it alone.

11.3 Chemical Potentials as Tangent Intercepts

Chemical potentials from $g(X)$. The Gibbs free energy of a phase is $G = N\,g(T,P,X)$ with $X = N_2/N$. At fixed (T,P), the derivatives $\partial X/\partial N_2|_{N_1} = (1-X)/N$ and $\partial X/\partial N_1|_{N_2} = -X/N$ give

$$\mu_2 = \left.\frac{\partial G}{\partial N_2}\right|_{T,P,N_1} = g + N\frac{\partial g}{\partial X}\frac{1-X}{N}, \qquad \mu_1 = \left.\frac{\partial G}{\partial N_1}\right|_{T,P,N_2} = g - N\frac{\partial g}{\partial X}\frac{X}{N},$$

that is,

$$\boxed{\mu_1 = g - X\,\frac{\partial g}{\partial X}, \qquad \mu_2 = g + (1-X)\,\frac{\partial g}{\partial X}.} \tag{11.2}$$

The tangent line to g at X therefore meets the axis $X = 0$ at μ_1 and the axis $X = 1$ at μ_2. This construction is the Legendre construction of the unary system with two intercepts instead of one. Substitution of Eq. (11.2) into Eq. (11.1) returns g, as required.

11.4 Two-Phase Equilibrium: Common Tangent and Lever Rule

Equal chemical potentials as a common tangent. Two phases, α and β, at the same (T,P) have separate curves $g^\alpha(X)$ and $g^\beta(X)$. Equilibrium requires equal chemical potentials for each component, so a binary system imposes two conditions, $\mu_1^\alpha = \mu_1^\beta$ and $\mu_2^\alpha = \mu_2^\beta$. The number of conditions equals the number of components, not the number of phases. By Eq. (11.2), the condition for component 1 equates the intercepts at $X = 0$ of the tangent lines at X^α and X^β, and the condition for component 2 equates the intercepts at $X = 1$. Two intercepts fix a straight line, so the two tangent lines coincide:

$$\boxed{g^{\alpha\prime}(X^\alpha) = g^{\beta\prime}(X^\beta) = \frac{g^\beta(X^\beta) - g^\alpha(X^\alpha)}{X^\beta - X^\alpha}.} \tag{11.3}$$

This condition is the common tangent construction (Fig. 8). The unary construction on $f(v)$ needed one intercept, because the unary system has one chemical potential. The binary construction fixes two intercepts, one for each component.

The equilibrium g is the convex hull. At fixed (T, P, N_1, N_2), the system minimizes G. An overall composition X between X^α and X^β can split into the two phases, and the free energy of this mixture lies on the straight segment of the common tangent. Where both single-phase curves lie above the segment, the two-phase mixture is stable. The equilibrium $g(X)$ is therefore the convex hull of all the single-phase curves. A composition on a curved part of the hull is a single phase, and a composition on a straight part splits into the two phases at the ends of the segment. The phase rule agrees, because at fixed (T, P), $F = 4 - \mathcal{P} - 2 = 0$ for two phases, so the compositions X^α and X^β are fixed.

The lever rule on a tie line. Here, $n_\alpha \equiv N^\alpha/N$ denotes the fraction of all particles in phase α, with $N^\alpha = N_1^\alpha + N_2^\alpha$. The segment between X^α and X^β at fixed (T, P) is the tie line of the two phases. Conservation of the particle number and of component 2 gives

$$n_\alpha + n_\beta = 1, \qquad n_\alpha X^\alpha + n_\beta X^\beta = X,$$

and hence the lever rule

$$\boxed{n_\alpha = \frac{X^\beta - X}{X^\beta - X^\alpha}, \qquad n_\beta = \frac{X - X^\alpha}{X^\beta - X^\alpha}.} \tag{11.4}$$

The rule is the unary lever rule with the composition X in place of the volume v.

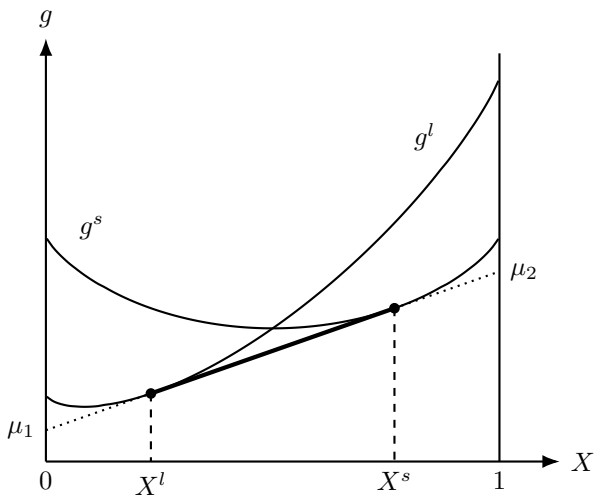


Figure 8: Common tangent construction for a solid (s) and a liquid (l) solution at fixed (T, P). Thick line: two-phase mixture on the convex hull. Intercepts: the common chemical potentials μ_1 and μ_2.

11.5 Three-Phase Equilibrium

One tangent line touching three curves. Here, the superscripts (1), (2) and (3) label three phases in order of composition, $X^{(1)} < X^{(2)} < X^{(3)}$. Three phases coexist when all three share the same μ_1 and μ_2. By Eq. (11.2), the three tangent lines then coincide, so the line $\lambda(X) =$

$(1-X)\mu_1 + X\mu_2$ touches all three curves:

$$\boxed{g^{(k)}\big(X^{(k)}\big) = \lambda\big(X^{(k)}\big), \qquad g^{(k)\prime}\big(X^{(k)}\big) = \mu_2 - \mu_1, \qquad k = 1, 2, 3.} \tag{11.5}$$

These six conditions contain the five unknowns μ_1, μ_2 and the three compositions, so at a fixed pressure they also fix the temperature. The phase rule agrees, because $F = 4-3 = 1$, and a fixed P leaves no freedom. In an isobaric T–X diagram, a three-phase equilibrium is therefore a horizontal line, the three-phase isotherm, with three fixed compositions. For an overall composition on this line, two conservation laws cannot fix three phase fractions, so the amounts of the three phases depend on the heat exchanged, not on the composition alone.

Which assemblage is stable above and below the isotherm. Let T_3 denote the temperature of the isotherm. The free energy change of forming the middle phase from the two outer phases at the fixed overall composition $X^{(2)}$ is

$$\Delta g(T) \equiv g^{(2)}\big(X^{(2)}\big) - n_1\, g^{(1)}\big(X^{(1)}\big) - n_3\, g^{(3)}\big(X^{(3)}\big), \qquad n_1 = \frac{X^{(3)} - X^{(2)}}{X^{(3)} - X^{(1)}}, \quad n_3 = 1 - n_1,$$

with $\Delta g(T_3) = 0$. The derivative of Δg contains the entropies $-s^{(k)} = \partial g^{(k)}/\partial T$ and terms from the temperature dependence of the compositions and of n_1. With the common slope $m = \mu_2 - \mu_1$ and the lever relation $n_1 X^{(1)} + n_3 X^{(3)} = X^{(2)}$, the composition terms sum to $m\,(X^{(1)} - X^{(3)})\,dn_1/dT$, and the terms from n_1 sum to $-(g^{(1)} - g^{(3)})\,dn_1/dT$. These two contributions cancel, because $m = (g^{(3)} - g^{(1)})/(X^{(3)} - X^{(1)})$. Hence

$$\boxed{\left.\frac{d\Delta g}{dT}\right|_{T_3} = -\Delta s_r, \qquad \Delta s_r \equiv s^{(2)} - n_1 s^{(1)} - n_3 s^{(3)}.} \tag{11.6}$$

If $\Delta s_r > 0$, the middle phase has the larger entropy, $\Delta g < 0$ above T_3, and the middle phase is stable above the isotherm. On cooling, it decomposes into the two outer phases. This case is the eutectic type, $(2) \to (1) + (3)$. If $\Delta s_r < 0$, the middle phase is stable below the isotherm and forms on cooling from the two outer phases. This case is the peritectic type, $(1) + (3) \to (2)$. In both cases, the reaction releases the latent heat $T_3|\Delta s_r|$ per particle on cooling. Three two-phase regions end on every isotherm, one for each pair of the three phases. For the eutectic type, two of them lie above the isotherm and one below. For the peritectic type, one lies above and two below.

Named three-phase reactions. Table 5 lists the three-phase reactions with established names. Other combinations of phases are possible, with the restriction that a system has only one gas phase, because all gases mix completely.

11.6 Shapes of Two-Phase Regions

Conjugate boundaries. At fixed pressure, a two-phase region is bounded by two conjugate curves, $X^{\alpha}(T)$ and $X^{\beta}(T)$, joined by horizontal tie lines. Both curves span the same temperature range. The region either closes where the two curves meet or ends on three-phase isotherms.

Table 5: Three-phase reactions on cooling. L: liquid. G: gas. Greek letters: solid phases.

Eutectic type, $(2) \to (1) + (3)$		Peritectic type, $(1) + (3) \to (2)$	
eutectic	$L \to \alpha + \beta$	peritectic	$L + \alpha \to \beta$
eutectoid	$\gamma \to \alpha + \beta$	peritectoid	$\alpha + \gamma \to \beta$
monotectic	$L_1 \to \alpha + L_2$	syntectic	$L_1 + L_2 \to \beta$
metatectic	$\beta \to \alpha + L$		$G + L \to \alpha$
	$L \to \alpha + G$		

Congruent points: the Gibbs–Konovalov relation. Along the two boundaries at fixed pressure, the Gibbs–Duhem relation per particle holds in each phase:

$$s^\alpha\,dT + (1 - X^\alpha)\,d\mu_1 + X^\alpha\,d\mu_2 = 0, \qquad s^\beta\,dT + (1 - X^\beta)\,d\mu_1 + X^\beta\,d\mu_2 = 0.$$

The difference of the two relations is the Gibbs–Konovalov relation

$$\boxed{(s^\alpha - s^\beta)\,dT = (X^\beta - X^\alpha)\,d(\mu_2 - \mu_1).} \tag{11.7}$$

By Eq. (11.2), $\mu_2 - \mu_1 = \partial g/\partial X$, which changes smoothly with the composition. Where the two phases have the same composition, $X^\alpha = X^\beta$ with $s^\alpha \neq s^\beta$, the coexistence temperature is therefore stationary, $dT/dX^\alpha = 0$. The two boundaries meet with horizontal tangents at a maximum or a minimum, called a congruent point, where the transformation occurs at a single temperature without a change of composition, as in a pure substance.

Critical points. Two phases with the same structure can also merge continuously. The two-phase region then closes at a critical point, where $X^\alpha = X^\beta$ and $s^\alpha = s^\beta$ hold together, and both sides of the Gibbs–Konovalov relation vanish. The miscibility gap of Section 11.9 closes in this way.

11.7 Solution Models

The ideal solution. Explicit phase diagrams require the composition dependence of each μ_i. The simplest assumption is that the two components just mix. An ideal solution is defined by

$$\boxed{\mu_i(T, P, X) = \mu_i^\circ(T, P) + k_B T \ln x_i, \qquad x_1 = 1 - X, \quad x_2 = X.} \tag{11.8}$$

The chemical potential of each component then lies below the pure value by $k_B T \ln x_i$. Equation (11.1) gives the Gibbs free energy per particle

$$\boxed{g_{\mathrm{id}}(T, P, X) = (1 - X)\,\mu_1^\circ + X\,\mu_2^\circ + k_B T\big[X \ln X + (1 - X)\ln(1 - X)\big].} \tag{11.9}$$

The regular solution. Here, w denotes the interaction parameter per particle. The regular solution adds to the ideal solution the simplest term that vanishes for both pure components and is symmetric in the composition:

$$\boxed{g_{\mathrm{reg}}(T, P, X) = g_{\mathrm{id}}(T, P, X) + w\,X(1 - X).} \tag{11.10}$$

The model favors unlike neighbors for $w < 0$ and like neighbors for $w > 0$. Section 11.9 shows that $w > 0$ can split one solution into two.

11.8 Isobaric T–X Diagram of an Isomorphous System

Isomorphous systems and the phase rule at fixed pressure. Two components are isomorphous when they dissolve completely in each other in both the liquid and the solid. This behavior requires the same crystal structure and similar atomic sizes, as in Cu–Ni, Si–Ge and Ag–Au. Most experiments on condensed phases run at a fixed pressure, so the phase diagram is the T–X slice of the X–P–T diagram. With P fixed, the phase rule becomes

$$\boxed{F = C + 1 - \mathcal{P} = 3 - \mathcal{P}.} \tag{11.11}$$

A single phase fills an area ($F = 2$), and two coexisting phases at a chosen T have fixed compositions ($F = 1$), which form a tie line.

Coexistence of ideal solid and liquid solutions. If both the solid and the liquid are ideal solutions, Eq. (11.8) turns the equilibrium condition $\mu_i^s = \mu_i^l$ into

$$\mu_i^{\circ,s} + k_B T \ln x_i^s = \mu_i^{\circ,l} + k_B T \ln x_i^l, \qquad \text{so} \qquad \frac{x_i^s}{x_i^l} = \exp\left(\frac{\mu_i^{\circ,l} - \mu_i^{\circ,s}}{k_B T}\right).$$

Pure component i melts at T_i with the latent heat per particle ℓ_i. For a constant ℓ_i, the entropy of fusion is ℓ_i/T_i, so $\mu_i^{\circ,l} - \mu_i^{\circ,s} = \ell_i(1 - T/T_i)$, and

$$\boxed{k_i(T) \equiv \frac{x_i^s}{x_i^l} = \exp\left[\frac{\ell_i}{k_B}\left(\frac{1}{T} - \frac{1}{T_i}\right)\right].} \tag{11.12}$$

The two conditions $x_1^s + x_2^s = 1$ and $x_1^l + x_2^l = 1$ then give $k_1(1 - X^l) + k_2 X^l = 1$, and therefore

$$\boxed{X^l(T) = \frac{1 - k_1}{k_2 - k_1}, \qquad X^s(T) = k_2\, X^l(T).} \tag{11.13}$$

The liquidus, the solidus and the lens. The curve $X^l(T)$ is the liquidus, above which the system is fully liquid. The curve $X^s(T)$ is the solidus, below which the system is fully solid. At $T = T_1$, $k_1 = 1$, and both curves reach $X = 0$. At $T = T_2$, $k_2 = 1$, and both reach $X = 1$. Between the two melting points, the curves enclose a lens-shaped two-phase region (Fig. 9). For $T_1 < T < T_2$, $k_2 > 1$, so $X^s > X^l$, and the solid is richer than the liquid in the component with the higher melting point.

Equilibrium cooling through the lens. A liquid of overall composition X_0 starts to solidify at the liquidus temperature T_L, where the first solid has the composition $X^s(T_L) > X_0$. On further cooling, the liquid composition follows the liquidus, the solid composition follows the solidus, and

the lever rule gives the amount of each phase on the tie line at every temperature. The last liquid disappears at the solidus temperature T_S, where the solid composition reaches X_0. A binary alloy therefore melts over the range $T_S < T < T_L$ instead of at a single temperature. At fixed X_0, the latent heat is released gradually over this range, so the entropy has no jump. Only the heat capacity jumps, at T_L and at T_S, unlike the single crease of a unary transition.

Non-equilibrium solidification. Equilibrium cooling requires the solid to change its composition along the solidus. Diffusion in a solid is slow, so in practice the first solid keeps the high content of component 2. The grains then have a composition gradient, called coring, and the remaining liquid becomes richer in component 1 than the diagram predicts. Zone refining uses the same separation to purify silicon, because each pass of a molten zone pushes the impurity into the liquid.

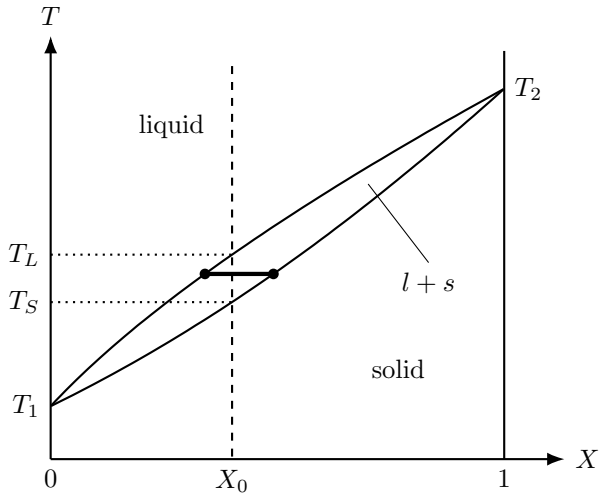


Figure 9: Isobaric T–X diagram of an isomorphous system with ideal solid and liquid solutions. Upper curve: liquidus. Lower curve: solidus. Thick line: tie line at one temperature during cooling of the composition X_0.

11.9 Miscibility Gap

Spinodal and critical temperature. For $w > 0$, the regular solution favors like pairs, and at low temperature the curve $g_{\mathrm{reg}}(X)$ becomes concave in a central range of compositions. Inside this range, a homogeneous solution is unstable against small composition fluctuations. The boundary is the spinodal, where the curvature vanishes:

$$\frac{\partial^2 g_{\mathrm{reg}}}{\partial X^2} = \frac{k_B T}{X(1-X)} - 2w = 0.$$

Hence

$$\boxed{T_{\mathrm{sp}}(X) = \frac{2w}{k_B}\, X(1-X), \qquad T_c = \frac{w}{2k_B}.} \tag{11.14}$$

The spinodal reaches the maximum, the critical temperature T_c, at $X = 1/2$. Above T_c, the components mix at every composition.

The binodal from the common tangent. Below T_c, the convex hull of the single curve $g_{\text{reg}}(X)$ has a common tangent that touches the curve at two compositions, and the solution splits into two solutions with the same structure. The linear terms of g do not affect the tangent points, and the remaining curve is symmetric about $X = 1/2$, so the common tangent is horizontal after the linear terms are removed. The tangent points therefore satisfy $k_B T \ln[X/(1-X)] + w(1-2X) = 0$, which gives the binodal

$$\boxed{T_{\text{b}}(X) = \frac{w}{k_B} \frac{1-2X}{\ln[(1-X)/X]}.} \tag{11.15}$$

The region under the binodal is the miscibility gap (Fig. 10). Between the binodal and the spinodal, a homogeneous solution is metastable, as a supercooled liquid is in the unary system. Inside the spinodal, the solution is unstable and separates spontaneously.

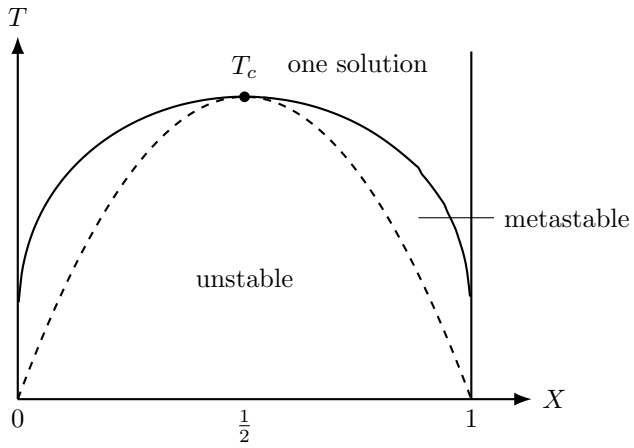


Figure 10: Miscibility gap of a regular solution. Solid curve: binodal (coexistence). Dashed curve: spinodal (limit of metastability).

11.10 Combining the Miscibility Gap with the Liquidus and Solidus

Computed diagrams. The common forms of isobaric diagrams follow from the convex hull of more than two curves $g^{\alpha}(X)$ at each temperature. Figure 11 shows eight diagrams computed in this way. In each panel, every boundary and every three-phase isotherm results from the minimization of G and is not drawn by hand. The models are the solution models of this section:

- The liquid is an ideal solution with $\mu_i^{\circ,l} - \mu_i^{\circ,s} = s_f(T_i - T)$, where T_i is the melting temperature of component i and s_f is the entropy of fusion per particle. In panel (f), the liquid is a regular solution.
- In panels (a) to (d) and (f), the solid is a single regular solution with parameter w_s, so the two terminal solids α and β have the same structure.

- In panels (e), (g) and (h), α and β have different structures and separate curves. In (g) and (h), an intermediate compound γ with a narrow curve near $X = 1/2$ is added.

Lens, maximum and minimum. For $w_s = 0$, the solid and the liquid are both ideal, and the diagram is the lens of Section 11.8, panel (a). For $w_s < 0$, unlike neighbors lower the solid curve at intermediate compositions, and the liquidus and the solidus meet at a congruent maximum, panel (b). For a small $w_s > 0$, the solid curve rises at intermediate compositions, and the liquidus and the solidus meet at a congruent minimum. The solid then has a miscibility gap with the critical temperature $w_s/2k_B$, which lies below the solidus and does not touch it, panel (c).

Eutectic from a gap that meets the solidus. A larger w_s raises the critical temperature of the gap until the gap meets the minimum of the solidus. The single solid region then splits into α and β, the congruent minimum becomes the eutectic point, and the liquid of this composition decomposes at one temperature into α and β, panel (d). The lower boundaries of α and β, called solvus curves, are the two branches of the gap.

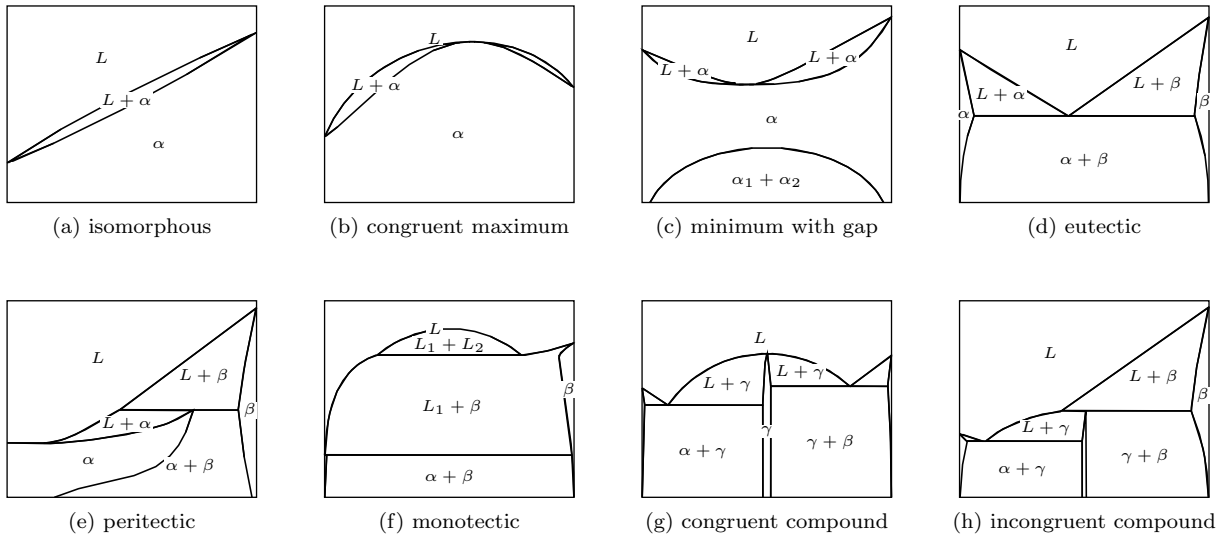
Peritectic from very different melting points. If the melting points of the two components differ strongly, the three-phase isotherm lies between them, and the liquid lies at one end of the isotherm instead of between the solids. Panel (e) shows the peritectic reaction $L + \beta \to \alpha$, in which α forms on cooling from the liquid and from the high-melting solid β.

Monotectic from a gap in the liquid. A regular liquid with $w_l > 0$ has a miscibility gap of its own. Where this gap meets the liquidus, a monotectic isotherm appears, on which one liquid decomposes on cooling into a solid and a second liquid, panel (f). At lower temperature, the remaining liquid freezes at a eutectic isotherm.

Intermediate phases. A compound γ that is stable up to the liquid melts congruently. The diagram then splits into two simple eutectic diagrams, A–γ and γ–B, panel (g). A less stable compound decomposes below its own melting point into a liquid and a solid. It then melts incongruently, by a peritectic reaction, panel (h).

11.11 Reading an Isobaric Diagram: Freezing Paths

The eutectic alloy. Here, X_0 denotes the overall composition of an alloy, and X_e, X^α and X^β denote the compositions of the liquid and of the two solids on the eutectic isotherm at T_e. An alloy of the eutectic composition stays liquid down to T_e and then freezes at this single temperature, as a pure substance does. Unlike a pure substance, the liquid and each solid differ in composition. The eutectic isotherm acts as a tie line between α and β, and the lever rule gives the fractions of



(a) isomorphous (b) congruent maximum (c) minimum with gap (d) eutectic

(e) peritectic (f) monotectic (g) congruent compound (h) incongruent compound

Figure 11: Isobaric T–X diagrams computed from the convex hull of the Gibbs free energy curves. Horizontal axis: X from 0 (pure A, left) to 1 (pure B, right). Vertical axis: T. Parameters, in units with $k_B = 1$: (a) $T_A = 1$, $T_B = 1.5$, $s_f = 1$, $w_s = 0$; (b)–(d) $T_A = 1$, $T_B = 1.15$, $s_f = 1.2$, $w_s = -1,\ 1.1,\ 2.2$; (e) $T_A = 1$, $T_B = 2$, separate solid structures; (f) $T_A = 0.6$, $T_B = 1.4$, $w_s = w_l = 3$; (g), (h) separate solid structures with a compound γ at $X = 1/2$. The generating scripts are in the folder `figures`.

the eutectic mixture,

$$\boxed{n_\alpha = \frac{X^\beta - X_e}{X^\beta - X^\alpha}, \qquad n_\beta = \frac{X_e - X^\alpha}{X^\beta - X^\alpha}.} \tag{11.16}$$

Hypoeutectic and hypereutectic alloys. An alloy with $X^\alpha < X_0 < X_e$, called hypoeutectic, starts to freeze at the liquidus. Between the liquidus and T_e, primary α forms, and the liquid composition follows the liquidus toward X_e. Just above T_e, the lever rule on the tie line from X^α to X_e gives the liquid fraction. At T_e, this liquid freezes into the eutectic mixture. The amounts after freezing are therefore

$$\boxed{n_{\text{eutectic}} = \frac{X_0 - X^\alpha}{X_e - X^\alpha}, \qquad n_{\text{primary }\alpha} = \frac{X_e - X_0}{X_e - X^\alpha}, \qquad n_\alpha^{\text{total}} = \frac{X^\beta - X_0}{X^\beta - X^\alpha}.} \tag{11.17}$$

A hypereutectic alloy, $X_e < X_0 < X^\beta$, behaves in the same way with primary β. An alloy outside the isotherm, $X_0 < X^\alpha$ or $X_0 > X^\beta$, freezes completely as one solid solution, as in the lens.

Peritectic alloys. For the peritectic reaction $L + \beta \to \alpha$ of Fig. 11(e), the compositions on the isotherm are ordered $X^L < X^\alpha < X^\beta$. Primary β forms above the isotherm. At the isotherm, the liquid reacts with β to form α. The overall composition decides which phase is used up first. For $X^\alpha < X_0 < X^\beta$, the liquid is used up, and $\alpha + \beta$ remains. For $X^L < X_0 < X^\alpha$, β is used up, and $L + \alpha$ remains and freezes further on cooling. For $X_0 = X^\alpha$, both are used up exactly, and a single α phase remains. The lever rule on the isotherm gives the amounts just above and just below T_p.

Non-equilibrium freezing. Equilibrium freezing requires each solid to change its composition by diffusion as the temperature falls. In practice, diffusion in solids is slow. Coring of the primary phase then leaves more liquid at the eutectic isotherm than the lever rule predicts, so even alloys outside the isotherm can contain some eutectic. In a peritectic reaction, the product forms a layer between the liquid and the primary solid, and further reaction requires diffusion through this layer. The reaction therefore often stops before completion.

Isothermal diffusion between two components. At fixed (T, P), both chemical potentials are constant across a two-phase mixture, because the coexisting phases share one tangent line. If diffusion fluxes are driven by gradients of the chemical potentials, no flux flows through a two-phase zone, and such a zone cannot grow. In a diffusion couple of A and B at fixed temperature, only one-phase layers form, in the order of the one-phase regions that the isotherm crosses in the diagram. Each two-phase region appears only as a sharp interface between two layers.

11.12 Assembly Rules for Isobaric Diagrams

Five rules. The phase rule and the preceding sections restrict how one-phase, two-phase and three-phase equilibria fit together in an isobaric binary diagram:

1. Two one-phase regions touch only at isolated congruent points, never along a line. By the Gibbs–Konovalov relation, equal compositions of two coexisting phases make the temperature stationary, which occurs only at isolated points.
2. Two neighboring one-phase regions are separated by the two-phase region of the same two phases.
3. Three two-phase regions end on every three-phase isotherm, one for each pair of the three phases, so six boundaries meet each isotherm.
4. Two three-phase isotherms are connected by a two-phase region only if they share two phases.
5. At a three-phase isotherm, the metastable extension of every two-phase boundary lies in a two-phase region, never in a one-phase region. This rule is the binary counterpart of the extension rule for the triple curves of the unary system.

A diagram from the univariant equilibria. The melting and polymorphic points of the components and of the congruent compounds, the three-phase isotherms and the critical points fix the arrangement of the two-phase regions. Usually, only one arrangement satisfies the five rules, and measurements then fix the exact positions of the boundaries.

11.13 Pressure: P–T Projections and Their Sections

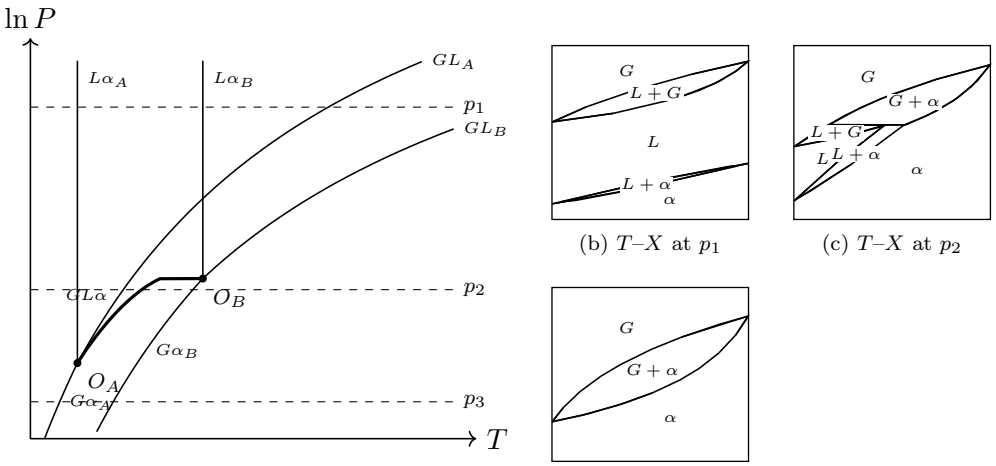
The gas phase. Isobaric diagrams of condensed systems usually omit the gas phase. A gas phase is present whenever a solid or liquid does not fill a sealed container, so the omission describes the

equilibrium incompletely. The omission is harmless when the pressure lies far above the vapor pressures of the condensed phases. Otherwise, the pressure must be included.

An isomorphous system with a gas phase. Here, ℓ_v denotes the latent heat of vaporization per particle, and $T_{b,i}$ the boiling temperature of pure i at the reference pressure $P = 1$. Figure 12 adds an ideal gas solution to the isomorphous model, with $\mu_i^{\circ,g} - \mu_i^{\circ,s} = s_f(T_i - T) + \ell_v(1 - T/T_{b,i}) + k_B T \ln P$. Each pure component contributes three triple curves in the P–T projection: sublimation ($G\alpha$), melting ($L\alpha$) and vaporization (GL), which meet at the triple points O_A and O_B. The binary three-phase curve $GL\alpha$ joins the two triple points. The isobaric sections follow from the same convex-hull construction:

- At p_1, above both triple points, the section contains two separate lenses, $L + \alpha$ for melting and $L + G$ for boiling.
- At p_2, between the triple points, the section crosses the $GL\alpha$ curve. The two lenses then end on a three-phase isotherm, on which the liquid decomposes on cooling into gas and solid, $L \to \alpha + G$, a eutectic-type reaction with a gas phase.
- At p_3, below both triple points, no liquid exists, and every alloy sublimes directly.

Isothermal P–X sections follow from the projection in the same way, with the positions of the fields reversed, because the dense phases are stable at high pressure but at low temperature.



(a) P–T projection

(b) T–X at p_1

(c) T–X at p_2

(d) T–X at p_3

Figure 12: Isomorphous system with solid (α), liquid (L) and gas (G) solutions, all ideal, computed from the convex hull. (a) P–T projection with the triple curves of the pure components and the binary three-phase curve $GL\alpha$. (b)–(d) Isobaric T–X sections at the pressures marked in (a). Parameters: $T_A = 1$, $T_B = 1.4$, $s_f = 1.2$, $\ell_v = 10$, $T_{b,A} = 1.8$, $T_{b,B} = 2.4$, with $k_B = 1$.

Quadruple points. In a system with a eutectic, the gas, the liquid and the two solids coexist at a quadruple point Q of the P–T projection (Fig. 13). Four three-phase curves, the quadruple curves,

meet at Q, one for each set of three of the four phases: $L\alpha\beta$, $G\alpha\beta$, $GL\alpha$ and $GL\beta$. At a quadruple point of the eutectic class, the $L\alpha\beta$ curve rises nearly vertically, the $G\alpha\beta$ curve runs toward lower temperature, and the $GL\alpha$ and $GL\beta$ curves run toward higher temperature, to the triple points of the components. Both components contribute to the vapor pressure of the $\alpha+\beta$ mixture, so the $G\alpha\beta$ curve lies above both sublimation curves. At a quadruple point of the peritectic class, the arrangement is reversed, with one curve toward higher and two toward lower temperature. Isobaric sections above Q give the ordinary eutectic diagram. Sections between Q and the triple points confine the liquid to a range of compositions, and sections below Q contain no liquid.

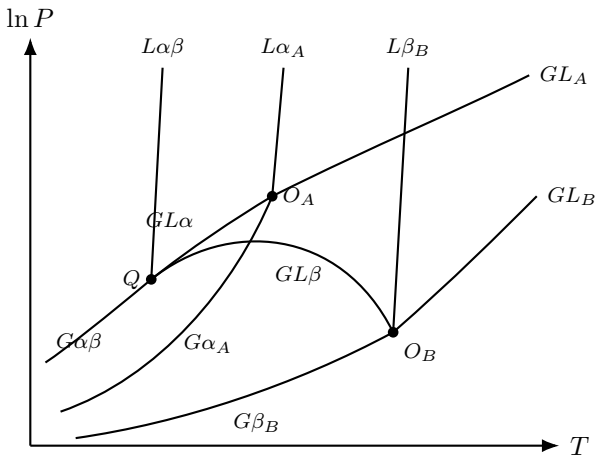


Figure 13: Schematic P–T projection of a binary eutectic system with the quadruple point Q, the four quadruple curves and the triple curves of the components.

11.14 Isothermal P–X Section of a Liquid–Vapor System

Raoult's law. Here, P_i denotes the partial pressure of component i in the vapor, $P_i^*(T)$ the vapor pressure of pure liquid i, and X^l and X^g the compositions of the liquid and the vapor. An ideal gas mixture has $\mu_i^g = \mu_i^{\circ,g}(T) + k_BT\ln P_i$, with $P_i = x_i^g P$. For the pure liquid in equilibrium with the vapor, $\mu_i^{\circ,l} = \mu_i^{\circ,g}(T) + k_BT\ln P_i^*$, where the weak pressure dependence of the liquid is neglected. An ideal liquid solution has $\mu_i^l = \mu_i^{\circ,l} + k_BT\ln x_i^l$, and the equality $\mu_i^l = \mu_i^g$ gives Raoult's law:

$$\boxed{P_i = x_i^l\, P_i^*.} \tag{11.18}$$

The bubble curve and the dew curve. The total pressure is the sum of the partial pressures,

$$\boxed{P = (1-X^l)\,P_1^* + X^l\,P_2^*,} \tag{11.19}$$

which is linear in X^l. This line is the bubble curve, where a compressed liquid forms the first bubble of vapor on decompression. The vapor composition follows from $X^g = P_2/P = X^lP_2^*/P$. Elimination of X^l with $1-X^l = (1-X^g)P/P_1^*$ and $X^l = X^gP/P_2^*$ gives the dew curve

$$\boxed{P = \left[\frac{1-X^g}{P_1^*} + \frac{X^g}{P_2^*}\right]^{-1},} \tag{11.20}$$

where the first drop of liquid forms on compression. The liquid is stable above the bubble curve, the vapor below the dew curve, and the two phases coexist between them (Fig. 14). The vapor is always richer in the more volatile component, the component with the larger P_i^*. The tie lines are horizontal, and the lever rule applies along them. The isobaric T–X diagram of the same system is the lens turned upside down, because the liquid is stable at low T but at high P.

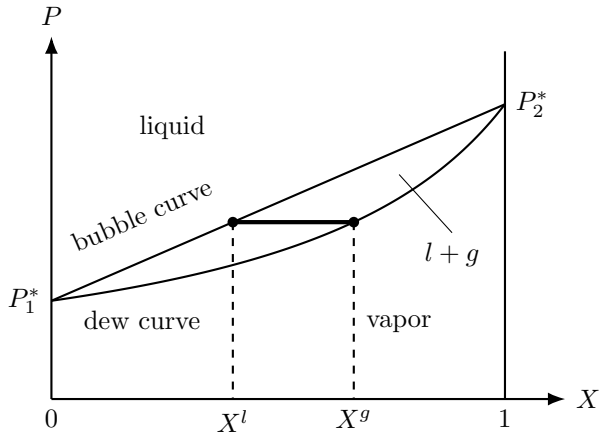


Figure 14: Isothermal P–X diagram of an ideal liquid–vapor system ($P_2^* = 3P_1^*$). Upper line: bubble curve. Lower curve: dew curve. Thick line: tie line.

Part IV

Physical Chemistry

Part V

Quantum Statistical Mechanics